1)

$I = \int \frac{1}{\sqrt{1-x^2}} \mathrm{d}x$

Use Integration by Substitution:

Let $x = \sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u$

$\Rightarrow \mathrm{d}x = \cos u \ \mathrm{d}u$

$\therefore I = \int \frac{1}{\sqrt{1-\sin^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{\cos u}{\sqrt{\cos^2 u}} \mathrm{d}u$

$= \int \frac{\cos u}{\cos u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \arcsin x + c$

2)

$I = \int \frac{1}{\sqrt{1-x^2}} \mathrm{d}x$

Use Integration by Substitution:

Let $x = \cos u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = -\sin u$

$\Rightarrow \mathrm{d}x = -\sin u \, \mathrm{d}u$

$\therefore I = \int \frac{1}{\sqrt{1-\cos^2 u}} \times -\sin u \, \mathrm{d}u$

$= -\int \frac{\sin u}{\sqrt{\sin^2 u}} \mathrm{d}u$

$= -\int \frac{\sin u}{\sin u} \mathrm{d}u$

$= -\int 1 \, \mathrm{d}u$

$= -u + c$

$= -\arccos x + c$

3)

$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$

$= [\arcsin x]_{-\frac{1}{2}}^{\frac{1}{2}}$

$= \left(\arcsin\left(\frac{1}{2}\right)\right) - \left(\arcsin\left(-\frac{1}{2}\right)\right)$

$= \left(\frac{\pi}{6}\right) - \left(-\frac{\pi}{6}\right)$

$= \frac{\pi}{3}$

4)

$I = \int \frac{1}{1+x^2}\,\mathrm{d}x$

Use Integration by Substitution:

Let $x = \tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u$

$\Rightarrow \mathrm{d}x = \sec^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{1}{1+\tan^2 u} \times \sec^2 u\,\mathrm{d}u$

$= \int \frac{\sec^2 u}{\sec^2 u}\,\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \arctan x + c$

5)

$\int_{-1}^{1} \frac{1}{1+x^2}\,\mathrm{d}x$

$= [\arctan x]_{-1}^{1}$

$= (\arctan 1) - (\arctan(-1))$

$= \left(\frac{\pi}{4}\right) - \left(-\frac{\pi}{4}\right)$

$= \frac{\pi}{2}$

6) a)

$\int \frac{2}{\sqrt{1-x^2}}\,\mathrm{d}x$

$= 2\arcsin x + c$

6) b)

$\int \frac{5}{1+x^2}\,\mathrm{d}x$

$= 5\arctan x + c$

6) c)

$\int \frac{1}{\sqrt{1-9x^2}}\,\mathrm{d}x$

$= \int \frac{1}{\sqrt{1-(3x)^2}}\,\mathrm{d}x$

$= \frac{1}{3}\arcsin 3x + c$

6) d)

$\int \frac{1}{1+4x^2}\,\mathrm{d}x$

$= \int \frac{1}{1+(2x)^2}\,\mathrm{d}x$

$= \frac{1}{2}\arctan 2x + c$

6) e)

$\int \frac{1}{\sqrt{1-2x^2}}\,\mathrm{d}x$

$= \int \frac{1}{\sqrt{1-\left(\sqrt{2}x\right)^2}}\,\mathrm{d}x$

$= \frac{1}{\sqrt{2}}\arcsin\left(\sqrt{2}x\right) + c$

$= \frac{\sqrt{2}}{2}\arcsin\left(\sqrt{2}x\right) + c$

6) f)

$\int \frac{1}{1+8x^2}\,\mathrm{d}x$

$= \int \frac{1}{1+\left(2\sqrt{2}x\right)^2}\,\mathrm{d}x$

$= \frac{1}{2\sqrt{2}}\arctan\left(2\sqrt{2}x\right) + c$

$= \frac{\sqrt{2}}{4}\arctan\left(2\sqrt{2}x\right) + c$

6) g)

$$\int \frac{10}{\sqrt{1-5x^2}}\,\mathrm{d}x$$
$$= \int \frac{10}{\sqrt{1-\left(\sqrt{5}x\right)^2}}\,\mathrm{d}x$$
$$= 10 \times \frac{1}{\sqrt{5}} \times \arcsin\left(\sqrt{5}x\right) + c$$
$$= 2\sqrt{5}\arcsin\left(\sqrt{5}x\right) + c$$

6) h)

$$\int \frac{200}{1+10x^2}\,\mathrm{d}x$$
$$= \int \frac{200}{1+\left(\sqrt{10}x\right)^2}\,\mathrm{d}x$$
$$= 100 \times \frac{1}{\sqrt{10}} \times \arctan\left(\sqrt{10}x\right) + c$$
$$= 20\sqrt{10}\arctan\left(\sqrt{10}x\right) + c$$

6) i)

$$\int \frac{4}{\sqrt{1-\frac{1}{4}x^2}}\,\mathrm{d}x$$
$$= \int \frac{4}{\sqrt{1-\left(\frac{1}{2}x\right)^2}}\,\mathrm{d}x$$
$$= 4 \times \frac{1}{\frac{1}{2}} \times \arcsin\left(\frac{1}{2}x\right) + c$$
$$= 8\arcsin\left(\frac{1}{2}x\right) + c$$

6) j)

$$\int \frac{18}{1+\frac{2}{3}x^2}\,\mathrm{d}x$$
$$= \int \frac{18}{1+\left(\sqrt{\frac{2}{3}}x\right)^2}\,\mathrm{d}x$$
$$= 18 \times \sqrt{\frac{2}{3}} \times \arctan\left(\sqrt{\frac{2}{3}}x\right) + c$$
$$= 9\sqrt{6}\arctan\left(\frac{\sqrt{6}}{3}x\right) + c$$

7) a)

$$\int_0^{\frac{\sqrt{2}}{8}} \frac{1}{\sqrt{1-16x^2}}\,\mathrm{d}x$$
$$= \int_0^{\frac{\sqrt{2}}{8}} \frac{1}{\sqrt{1-(4x)^2}}\,\mathrm{d}x$$
$$= \left[\frac{1}{4}\arcsin(4x)\right]_0^{\frac{\sqrt{2}}{8}}$$
$$= \left(\frac{1}{4}\arcsin\left(4\times\frac{\sqrt{2}}{8}\right)\right) - \left(\frac{1}{4}\arcsin(4\times 0)\right)$$
$$= \left(\frac{\pi}{16}\right) - (0)$$
$$= \frac{\pi}{16}$$

7) b)

$$\int_{\frac{1}{2\sqrt{3}}}^{\frac{1}{2}} \frac{30}{\sqrt{1-3x^2}}\,\mathrm{d}x$$
$$= \int_{\frac{1}{2\sqrt{3}}}^{\frac{1}{2}} \frac{30}{\sqrt{1-\left(\sqrt{3}x\right)^2}}\,\mathrm{d}x$$
$$= \left[30\times\frac{1}{\sqrt{3}}\times\arcsin\left(\sqrt{3}x\right)\right]_{\frac{1}{2\sqrt{3}}}^{\frac{1}{2}}$$
$$= \left[10\sqrt{3}\arcsin\left(\sqrt{3}x\right)\right]_{\frac{1}{2\sqrt{3}}}^{\frac{1}{2}}$$
$$= 10\sqrt{3}\arcsin\frac{\sqrt{3}}{2} - 10\sqrt{3}\arcsin\frac{1}{2}$$
$$= 10\sqrt{3}\times\frac{\pi}{3} - 10\sqrt{3}\times\frac{\pi}{6}$$
$$= \frac{5\sqrt{3}\pi}{3}$$

8) a)

$$\int_0^{\frac{\sqrt{3}}{6}} \frac{1}{1+36x^2}\,\mathrm{d}x$$
$$= \int_0^{\frac{\sqrt{3}}{6}} \frac{1}{1+(6x)^2}\,\mathrm{d}x$$
$$= \left[\frac{1}{6}\arctan(6x)\right]_0^{\frac{\sqrt{3}}{6}}$$
$$= \left(\frac{1}{6}\arctan\left(6\times\frac{\sqrt{3}}{6}\right)\right) - \left(\frac{1}{6}\arctan(6\times 0)\right)$$
$$= \left(\frac{\pi}{18}\right) - (0)$$
$$= \frac{\pi}{18}$$

8) b)

$$\int_{\frac{1}{\sqrt{3}}}^{1} \frac{12}{1+3x^2}\,\mathrm{d}x$$
$$= \int_{\frac{1}{\sqrt{3}}}^{1} \frac{12}{1+\left(\sqrt{3}x\right)^2}\,\mathrm{d}x$$
$$= \left[12\times\frac{1}{\sqrt{3}}\times\arctan\left(\sqrt{3}x\right)\right]_{\frac{1}{\sqrt{3}}}^{1}$$
$$= \left[4\sqrt{3}\arctan\left(\sqrt{3}x\right)\right]_{\frac{1}{\sqrt{3}}}^{1}$$
$$= 4\sqrt{3}\arctan\sqrt{3} - 4\sqrt{3}\arctan 1$$
$$= 4\sqrt{3}\times\frac{\pi}{3} - 4\sqrt{3}\times\frac{\pi}{4}$$
$$= \frac{\sqrt{3}\pi}{3}$$

9) a)

$$\frac{1}{2+x^2} \equiv \frac{1}{2\left(1+\frac{1}{2}x^2\right)} \equiv \frac{1}{2\left(1+\left(\frac{1}{\sqrt{2}}x\right)^2\right)}$$

So $k = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

9) b)

$$\int \frac{1}{2+x^2}\,\mathrm{d}x = \int \frac{1}{2\left(1+\left(\frac{1}{\sqrt{2}}x\right)^2\right)}\,\mathrm{d}x$$

$$= \frac{1}{2}\int \frac{1}{1+\left(\frac{1}{\sqrt{2}}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{2} \times \frac{1}{\frac{1}{\sqrt{2}}} \times \arctan\left(\frac{1}{\sqrt{2}}x\right) + c$$

$$= \frac{\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{2}x\right) + c$$

10) a)

$$\frac{1}{\sqrt{2-x^2}} \equiv \frac{1}{\sqrt{2\left(1-\frac{1}{2}x^2\right)}} \equiv \frac{1}{\sqrt{2}\sqrt{1-\left(\frac{1}{\sqrt{2}}x\right)^2}}$$

So $k = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

10) b)

$$\int \frac{1}{\sqrt{2-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{2}\sqrt{1-\left(\frac{1}{\sqrt{2}}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{\sqrt{2}}\int \frac{1}{\sqrt{1-\left(\frac{1}{\sqrt{2}}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\frac{1}{\sqrt{2}}} \times \arcsin\left(\frac{1}{\sqrt{2}}x\right) + c$$

$$= \arcsin\left(\frac{\sqrt{2}}{2}x\right) + c$$

11) a)

$$\int \frac{1}{4+x^2}\,\mathrm{d}x = \int \frac{1}{4\left(1+\frac{1}{4}x^2\right)}\,\mathrm{d}x = \frac{1}{4}\int \frac{1}{1+\left(\frac{1}{2}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{4} \times \frac{1}{\frac{1}{2}}\arctan\left(\frac{1}{2}x\right) + c$$

$$= \frac{1}{2}\arctan\left(\frac{1}{2}x\right) + c$$

11) b)

$$\int \frac{1}{\sqrt{9-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{9\left(1-\frac{1}{9}x^2\right)}}\,\mathrm{d}x = \frac{1}{3}\int \frac{1}{\sqrt{1-\left(\frac{1}{3}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{3} \times \frac{1}{\frac{1}{3}} \arcsin\left(\frac{1}{3}x\right) + c$$

$$= \arcsin\left(\frac{1}{3}x\right) + c$$

11) c)

$$\int \frac{1}{5+x^2}\,\mathrm{d}x = \int \frac{1}{5\left(1+\frac{1}{5}x^2\right)}\,\mathrm{d}x = \frac{1}{5}\int \frac{1}{1+\left(\frac{1}{\sqrt{5}}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{5} \times \frac{1}{\frac{1}{\sqrt{5}}} \arctan\left(\frac{1}{\sqrt{5}}x\right) + c$$

$$= \frac{\sqrt{5}}{5} \arctan\left(\frac{\sqrt{5}}{5}x\right) + c$$

11) d)

$$\int \frac{1}{\sqrt{8-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{8\left(1-\frac{1}{8}x^2\right)}}\,\mathrm{d}x = \frac{1}{2\sqrt{2}}\int \frac{1}{\sqrt{1-\left(\frac{1}{2\sqrt{2}}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{2\sqrt{2}} \times \frac{1}{\frac{1}{2\sqrt{2}}} \arcsin\left(\frac{1}{2\sqrt{2}}x\right) + c$$

$$= \arcsin\left(\frac{\sqrt{2}}{4}x\right) + c$$

11) e)

$$\int \frac{4}{12+x^2}\,\mathrm{d}x = \int \frac{4}{12\left(1+\frac{1}{12}x^2\right)}\,\mathrm{d}x = \frac{1}{3}\int \frac{1}{1+\left(\frac{1}{2\sqrt{3}}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{3} \times \frac{1}{\frac{1}{2\sqrt{3}}} \arctan\left(\frac{1}{2\sqrt{3}}x\right) + c$$

$$= \frac{2\sqrt{3}}{3} \arctan\left(\frac{\sqrt{3}}{6}x\right) + c$$

11) f)

$$\int \frac{10}{\sqrt{5-x^2}}\,\mathrm{d}x = \int \frac{10}{\sqrt{5\left(1-\frac{1}{5}x^2\right)}}\,\mathrm{d}x = \frac{10}{\sqrt{5}}\int \frac{1}{\sqrt{1-\left(\frac{1}{\sqrt{5}}x\right)^2}}\,\mathrm{d}x$$

$$= 2\sqrt{5} \times \frac{1}{\frac{1}{\sqrt{5}}} \arcsin\left(\frac{1}{\sqrt{5}}x\right) + c$$

$$= 10 \arcsin\left(\frac{\sqrt{5}}{5}x\right) + c$$

11) g)

$$\int \frac{1}{2+4x^2}\,\mathrm{d}x = \int \frac{1}{2(1+2x^2)}\,\mathrm{d}x = \frac{1}{2}\int \frac{1}{1+\left(\sqrt{2}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} \arctan\left(\sqrt{2}x\right) + c$$

$$= \frac{\sqrt{2}}{4} \arctan\left(\sqrt{2}x\right) + c$$

11) h)

$$\int \frac{1}{\sqrt{16-9x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{16\left(1-\frac{9}{16}x^2\right)}}\,\mathrm{d}x = \frac{1}{4}\int \frac{1}{\sqrt{1-\left(\frac{3}{4}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{4} \times \frac{1}{\frac{3}{4}} \arcsin\left(\frac{3}{4}x\right) + c$$

$$= \frac{1}{3} \arcsin\left(\frac{3}{4}x\right) + c$$

11) i)

$$\int \frac{2}{4+25x^2}\,\mathrm{d}x = \int \frac{2}{4\left(1+\frac{25}{4}x^2\right)}\,\mathrm{d}x = \frac{1}{2}\int \frac{1}{1+\left(\frac{5}{2}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{2} \times \frac{1}{\frac{5}{2}} \arctan\left(\frac{5}{2}x\right) + c$$

$$= \frac{1}{5} \arctan\left(\frac{5}{2}x\right) + c$$

11) j)

$$\int \frac{3}{\sqrt{12-\frac{1}{3}x^2}}\,\mathrm{d}x = \int \frac{3}{\sqrt{12\left(1-\frac{1}{36}x^2\right)}}\,\mathrm{d}x = \frac{\sqrt{3}}{2}\int \frac{1}{\sqrt{1-\left(\frac{1}{6}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\frac{1}{6}} \arcsin\left(\frac{1}{6}x\right) + c$$

$$= 3\sqrt{3} \arcsin\left(\frac{1}{6}x\right) + c$$

12) a)

$$\int_{-\frac{\sqrt{3}}{3}}^{\frac{\sqrt{3}}{3}} \frac{1}{\sqrt{4-9x^2}}\,\mathrm{d}x$$

$$= \int_{-\frac{\sqrt{3}}{3}}^{\frac{\sqrt{3}}{3}} \frac{1}{\sqrt{4\left(1-\frac{9}{4}x^2\right)}}\,\mathrm{d}x$$

$$= \frac{1}{2}\int_{-\frac{\sqrt{3}}{3}}^{\frac{\sqrt{3}}{3}} \frac{1}{\sqrt{1-\left(\frac{3}{2}x\right)^2}}\,\mathrm{d}x$$

$$= \left[\frac{1}{2} \times \frac{1}{\frac{3}{2}} \arcsin\left(\frac{3}{2}x\right)\right]_{-\frac{\sqrt{3}}{3}}^{\frac{\sqrt{3}}{3}}$$

$$= \left[\frac{1}{3}\arcsin\left(\frac{3}{2}x\right)\right]_{-\frac{\sqrt{3}}{3}}^{\frac{\sqrt{3}}{3}}$$

$$= \left(\frac{1}{3}\arcsin\left(\frac{3}{2} \times \frac{\sqrt{3}}{3}\right)\right) - \left(\frac{1}{3}\arcsin\left(\frac{3}{2} \times \left(-\frac{\sqrt{3}}{3}\right)\right)\right)$$

$$= \left(\frac{\pi}{9}\right) - \left(-\frac{\pi}{9}\right)$$

$$= \frac{2\pi}{9}$$

12) b)

$$\int_0^{\frac{5}{8}(\sqrt{5}-1)} \frac{3}{\sqrt{100-16x^2}}\,\mathrm{d}x$$

$$= \int_0^{\frac{5}{8}(\sqrt{5}-1)} \frac{3}{\sqrt{100\left(1-\frac{16}{100}x^2\right)}}\,\mathrm{d}x$$

$$= \frac{3}{10}\int_0^{\frac{5}{8}(\sqrt{5}-1)} \frac{1}{\sqrt{1-\left(\frac{4}{10}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{3}{10}\int_0^{\frac{5}{8}(\sqrt{5}-1)} \frac{1}{\sqrt{1-\left(\frac{2}{5}x\right)^2}}\,\mathrm{d}x$$

$$= \left[\frac{3}{10} \times \frac{1}{\frac{2}{5}} \arcsin\left(\frac{2}{5}x\right)\right]_0^{\frac{5}{8}(\sqrt{5}-1)}$$

$$= \left[\frac{3}{4}\arcsin\left(\frac{2}{5}x\right)\right]_0^{\frac{5}{8}(\sqrt{5}-1)}$$

$$= \left(\frac{3}{4}\arcsin\left(\frac{2}{5} \times \frac{5}{8}\left(\sqrt{5}-1\right)\right)\right) - \left(\frac{3}{4}\arcsin\left(\frac{2}{5} \times 0\right)\right)$$

$$= \left(\frac{3\pi}{40}\right) - (0)$$

$$= \frac{3\pi}{40}$$

13) a)

$$\int_{-\frac{5\sqrt{3}}{27}}^{\frac{5\sqrt{3}}{27}} \frac{1}{25+81x^2}\,\mathrm{d}x$$

$$= \int_{-\frac{5\sqrt{3}}{27}}^{\frac{5\sqrt{3}}{27}} \frac{1}{25\left(1+\frac{81}{25}x^2\right)}\,\mathrm{d}x$$

$$= \frac{1}{25}\int_{-\frac{5\sqrt{3}}{27}}^{\frac{5\sqrt{3}}{27}} \frac{1}{1+\left(\frac{9}{5}x\right)^2}\,\mathrm{d}x$$

$$= \left[\frac{1}{25} \times \frac{1}{\frac{9}{5}} \arctan\left(\frac{9}{5}x\right)\right]_{-\frac{5\sqrt{3}}{27}}^{\frac{5\sqrt{3}}{27}}$$

$$= \left[\frac{1}{45}\arctan\left(\frac{9}{5}x\right)\right]_{-\frac{5\sqrt{3}}{27}}^{\frac{5\sqrt{3}}{27}}$$

$$= \left(\frac{1}{45}\arctan\left(\frac{9}{5} \times \frac{5\sqrt{3}}{27}\right)\right) - \left(\frac{1}{45}\arctan\left(\frac{9}{5} \times \left(-\frac{5\sqrt{3}}{27}\right)\right)\right)$$

$$= \left(\frac{\pi}{270}\right) - \left(-\frac{\pi}{270}\right)$$

$$= \frac{\pi}{135}$$

13) b)

$$\int_0^{3\left(\sqrt{2}-\sqrt{\frac{3}{2}}\right)} \frac{9}{9+2x^2}\,\mathrm{d}x$$

$$= \int_0^{3\left(\sqrt{2}-\sqrt{\frac{3}{2}}\right)} \frac{9}{9\left(1+\frac{2}{9}x^2\right)}\,\mathrm{d}x$$

$$= \int_0^{3\left(\sqrt{2}-\sqrt{\frac{3}{2}}\right)} \frac{1}{1+\left(\frac{\sqrt{2}}{3}x\right)^2}\,\mathrm{d}x$$

$$= \left[\frac{1}{\frac{\sqrt{2}}{3}}\arctan\left(\frac{\sqrt{2}}{3}x\right)\right]_0^{3\left(\sqrt{2}-\sqrt{\frac{3}{2}}\right)}$$

$$= \left[\frac{3\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{3}x\right)\right]_0^{3\left(\sqrt{2}-\sqrt{\frac{3}{2}}\right)}$$

$$= \left(\frac{3\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{3}\times 3\left(\sqrt{2}-\sqrt{\frac{3}{2}}\right)\right)\right) - \left(\frac{3\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}}{3}\times 0\right)\right)$$

$$= \left(\frac{\sqrt{2}}{8}\pi\right) - (0)$$

$$= \frac{\pi\sqrt{2}}{8}$$

14) a)

$$\int \frac{1}{\sqrt{a^2-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{a^2\left(1-\frac{1}{a^2}x^2\right)}}\,\mathrm{d}x = \frac{1}{a}\int \frac{1}{\sqrt{1-\left(\frac{1}{a}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{a}\times\frac{1}{\frac{1}{a}}\arcsin\left(\frac{1}{a}x\right)+c$$

$$= \arcsin\left(\frac{x}{a}\right)+c$$

14) b)

$$\int \frac{1}{a^2+x^2}\,\mathrm{d}x = \int \frac{1}{a^2\left(1+\frac{1}{a^2}x^2\right)}\,\mathrm{d}x = \frac{1}{a^2}\int \frac{1}{1+\left(\frac{1}{a}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{a^2}\times\frac{1}{\frac{1}{a}}\arctan\left(\frac{1}{a}x\right)+c$$

$$= \frac{1}{a}\arctan\left(\frac{x}{a}\right)+c$$

14) c)

$$\int \frac{1}{\sqrt{a^2-b^2x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{a^2\left(1-\frac{b^2}{a^2}x^2\right)}}\,\mathrm{d}x = \frac{1}{a}\int \frac{1}{\sqrt{1-\left(\frac{b}{a}x\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{a} \times \frac{1}{\frac{b}{a}} \arcsin\left(\frac{b}{a}x\right) + c$$

$$= \frac{1}{b} \arcsin\left(\frac{b}{a}x\right) + c$$

14) d)

$$\int \frac{1}{a^2+b^2x^2}\,\mathrm{d}x = \int \frac{1}{a^2\left(1+\frac{b^2}{a^2}x^2\right)}\,\mathrm{d}x = \frac{1}{a^2}\int \frac{1}{1+\left(\frac{b}{a}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{a^2} \times \frac{1}{\frac{b}{a}} \arctan\left(\frac{b}{a}x\right) + c$$

$$= \frac{1}{ab} \arctan\left(\frac{b}{a}x\right) + c$$

15)

$$\int_0^{\sqrt{ab}} \frac{1}{\sqrt{b-\frac{x^2}{a}}}\,\mathrm{d}x$$

$$= \int_0^{\sqrt{ab}} \frac{1}{\sqrt{b\left(1-\frac{x^2}{ab}\right)}}\,\mathrm{d}x$$

$$= \frac{1}{\sqrt{b}}\int_0^{\sqrt{ab}} \frac{1}{\sqrt{1-\left(\frac{x}{\sqrt{ab}}\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{\sqrt{b}}\left[\sqrt{ab} \times \arcsin\left(\frac{x}{\sqrt{ab}}\right)\right]_0^{\sqrt{ab}}$$

$$= \left[\sqrt{a} \arcsin\left(\frac{x}{\sqrt{ab}}\right)\right]_0^{\sqrt{ab}}$$

$$= \sqrt{a} \arcsin\left(\frac{\sqrt{ab}}{\sqrt{ab}}\right) - \sqrt{a} \arcsin\left(\frac{0}{\sqrt{ab}}\right)$$

$$= \sqrt{a} \arcsin 1 - \sqrt{a} \arcsin 0$$

$$= \sqrt{a} \times \frac{\pi}{2} - 0$$

$$= \frac{\pi\sqrt{a}}{2}$$

16)

$$\int_{\frac{\sqrt{6}}{2}}^{\sqrt{3}} \frac{6}{\sqrt{18-3x^2}}\,\mathrm{d}x$$

$$= \int_{\frac{\sqrt{6}}{2}}^{\sqrt{3}} \frac{6}{\sqrt{18\left(1-\frac{1}{6}x^2\right)}}\,\mathrm{d}x$$

$$= \sqrt{2}\int_{\frac{\sqrt{6}}{2}}^{\sqrt{3}} \frac{1}{\sqrt{1-\left(\frac{1}{\sqrt{6}}x\right)^2}}\,\mathrm{d}x$$

$$= \left[\sqrt{2} \times \frac{1}{\frac{1}{\sqrt{6}}} \arcsin\left(\frac{1}{\sqrt{6}}x\right)\right]_{\frac{\sqrt{6}}{2}}^{\sqrt{3}}$$

$$= \left[2\sqrt{3}\arcsin\left(\frac{\sqrt{6}}{6}x\right)\right]_{\frac{\sqrt{6}}{2}}^{\sqrt{3}}$$

$$= \left(2\sqrt{3}\arcsin\left(\frac{\sqrt{6}}{6}\times\sqrt{3}\right)\right) - \left(2\sqrt{3}\arcsin\left(\frac{\sqrt{6}}{6}\times\frac{\sqrt{6}}{2}\right)\right)$$

$$= \left(2\sqrt{3}\arcsin\frac{\sqrt{2}}{2}\right) - \left(2\sqrt{3}\arcsin\frac{1}{2}\right)$$

$$= \left(\frac{\sqrt{3}}{2}\pi\right) - \left(\frac{\sqrt{3}}{3}\pi\right)$$

$$= \frac{\sqrt{3}}{6}\pi$$

$$= \frac{\pi\sqrt{3}}{6}\ \text{units}^2$$

17)

$$\int_{3\sqrt{3}-6}^{6-3\sqrt{3}} \frac{9}{36+4x^2}\,\mathrm{d}x$$

$$= \int_{3\sqrt{3}-6}^{6-3\sqrt{3}} \frac{9}{36\left(1+\frac{1}{9}x^2\right)}\,\mathrm{d}x$$

$$= \frac{1}{4}\int_{3\sqrt{3}-6}^{6-3\sqrt{3}} \frac{1}{1+\frac{1}{9}x^2}\,\mathrm{d}x$$

$$= \frac{1}{4}\int_{3\sqrt{3}-6}^{6-3\sqrt{3}} \frac{1}{1+\left(\frac{1}{3}x\right)^2}\,\mathrm{d}x$$

$$= \left[\frac{1}{4}\times\frac{1}{\frac{1}{3}}\arctan\left(\frac{1}{3}x\right)\right]_{3\sqrt{3}-6}^{6-3\sqrt{3}}$$

$$= \left[\frac{3}{4}\arctan\left(\frac{1}{3}x\right)\right]_{3\sqrt{3}-6}^{6-3\sqrt{3}}$$

$$= \left(\frac{3}{4}\arctan\left(\frac{1}{3}\times\left(6-3\sqrt{3}\right)\right)\right) - \left(\frac{3}{4}\arctan\left(\frac{1}{3}\times\left(3\sqrt{3}-6\right)\right)\right)$$

$$= \left(\frac{\pi}{16}\right) - \left(-\frac{\pi}{16}\right)$$

$$= \frac{\pi}{8}\ \text{units}^2$$

18) a)

$$\frac{19x^2+34}{(1+x^2)(9+4x^2)} \equiv \frac{Ax+B}{1+x^2} + \frac{Cx+D}{9+4x^2}$$

$$\Rightarrow 19x^2+34 \equiv (Ax+B)(9+4x^2)+(Cx+D)(1+x^2)$$

Let $x=0$: $34 = 9B+D$

Let $x=1$: $53 = 13A+13B+2C+2D$

Let $x=-1$: $53 = -13A+13B-2C+2D$

Let $x=2$: $110 = 50A+25B+10C+5D$

$A=0,\ B=3,\ C=0,\ D=7$

$$\therefore \frac{19x^2+34}{(1+x^2)(9+4x^2)} \equiv \frac{3}{1+x^2} + \frac{7}{9+4x^2}$$

18) b)

$$\int \frac{19x^2+34}{(1+x^2)(9+4x^2)}\,\mathrm{d}x = \int \left(\frac{3}{1+x^2}+\frac{7}{9+4x^2}\right)\mathrm{d}x$$

$$= \int \left(\frac{3}{1+x^2}+\frac{7}{9\left(1+\frac{4}{9}x^2\right)}\right)\mathrm{d}x$$

$$= \int \left(\frac{3}{1+x^2}+\frac{7}{9\left(1+\left(\frac{2}{3}x\right)^2\right)}\right)\mathrm{d}x$$

$$= 3\arctan x + \frac{7}{9}\times\frac{1}{\frac{2}{3}}\times\arctan\left(\frac{2}{3}x\right)+c$$

$$= 3\arctan x + \frac{7}{6}\arctan\left(\frac{2}{3}x\right)+c$$

19) a)

$$\frac{107x^2-3}{(4+9x^2)(1+25x^2)} \equiv \frac{Ax+B}{4+9x^2}+\frac{Cx+D}{1+25x^2}$$

$$\Rightarrow 107x^2-3 \equiv (Ax+B)(1+25x^2)+(Cx+D)(4+9x^2)$$

Let $x=0$: $-3 = B+4D$

Let $x=1$: $104 = 26A+26B+13C+13D$

Let $x=-1$: $104 = -26A+26B-13C+13D$

Let $x=2$: $425 = 202A+101B+80C+40D$

$A=0,\ B=5,\ C=0,\ D=-2$

$$\therefore \frac{107x^2-3}{(4+9x^2)(1+25x^2)} \equiv \frac{5}{4+9x^2}-\frac{2}{1+25x^2}$$

$$\int \frac{107x^2-3}{(4+9x^2)(1+25x^2)}\,\mathrm{d}x = \int \left(\frac{5}{4+9x^2}-\frac{2}{1+25x^2}\right)\mathrm{d}x$$

$$= \int \left(\frac{5}{4\left(1+\frac{9}{4}x^2\right)}-\frac{2}{1+(5x)^2}\right)\mathrm{d}x$$

$$= \int \left(\frac{5}{4\left(1+\left(\frac{3}{2}x\right)^2\right)}-\frac{2}{1+(5x)^2}\right)\mathrm{d}x$$

$$= \frac{5}{4}\times\frac{1}{\frac{3}{2}}\times\arctan\left(\frac{3}{2}x\right)-2\times\frac{1}{5}\times\arctan 5x+c$$

$$= \frac{5}{6}\arctan\left(\frac{3}{2}x\right)-\frac{2}{5}\arctan 5x+c$$

19) b)

$\frac{8x^4+12x^3+33x^2+3x+4}{(4+x^2)(1+4x^2)} \equiv E + \frac{Ax+B}{4+x^2} + \frac{Cx+D}{1+4x^2}$

The coefficient of x^4 in the numerator is 8 and the coefficient of x^4 in the denominator is 4, so $E = \frac{8}{4} = 2$

$\therefore \frac{8x^4+12x^3+33x^2+3x+4}{(4+x^2)(1+4x^2)} \equiv 2 + \frac{Ax+B}{4+x^2} + \frac{Cx+D}{1+4x^2}$

$\Rightarrow 8x^4 + 12x^3 + 33x^2 + 3x + 4 \equiv 2(4 + x^2)(1 + 4x^2) + (Ax + B)(1 + 4x^2) + (Cx + D)(4 + x^2)$

Let $x = 0$: $4 = 8 + B + 4D \Rightarrow B + D = -4$

Let $x = 1$: $60 = 50 + 5A + 5B + 5C + 5D \Rightarrow A + B + C + D = 2$

Let $x = -1$: $30 = 50 - 5A + 5B - 5C + 5D \Rightarrow A - B + C - D = 4$

Let $x = 2$: $366 = 272 + 34A + 17B + 16C + 8D \Rightarrow 34A + 17B + 16C + 8D = 94$

$A = 3$, $B = 0$, $C = 0$, $D = -1$

$\therefore \frac{8x^4+12x^3+33x^2+3x+4}{(4+x^2)(1+4x^2)} \equiv 2 + \frac{3x}{4+x^2} - \frac{1}{1+4x^2}$

$\int \frac{8x^4+12x^3+33x^2+3x+4}{(4+x^2)(1+4x^2)} \mathrm{d}x = \int \left(2 + \frac{3x}{4+x^2} - \frac{1}{1+4x^2}\right) \mathrm{d}x$

$= \int \left(2 + \frac{3x}{4+x^2} - \frac{1}{1+(2x)^2}\right) \mathrm{d}x$

$= 2x + \frac{3}{2}\ln|4 + x^2| - \frac{1}{2}\arctan 2x + c$

20)

$\frac{16x^2+5x+4}{x(8x^2+2)} \equiv \frac{A}{x} + \frac{Bx+C}{8x^2+2}$

$\Rightarrow 16x^2 + 5x + 4 \equiv A(8x^2 + 2) + (Bx + C)x$

Let $x = 0$: $4 = 2A \Rightarrow A = 2$

Let $x = 1$: $25 = 20 + B + C \Rightarrow B + C = 5$

Let $x = -1$: $15 = 20 + B - C \Rightarrow B - C = -5$

$B = 0$, $C = 5$

$\therefore \frac{16x^2+5x+4}{x(8x^2+2)} \equiv \frac{2}{x} + \frac{5}{8x^2+2}$

So:

$\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{16x^2+5x+4}{x(8x^2+2)} \mathrm{d}x = \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \left(\frac{2}{x} + \frac{5}{8x^2+2}\right) \mathrm{d}x$

$= \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \left(\frac{2}{x} + \frac{5}{2(1+4x^2)}\right) \mathrm{d}x$

$= \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \left(\frac{2}{x} + \frac{5}{2(1+(2x)^2)}\right) \mathrm{d}x$

$= \left[2\ln|x| + \frac{5}{2} \times \frac{1}{2} \times \arctan 2x\right]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$

$= \left[2\ln|x| + \frac{5}{4}\arctan 2x\right]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$

$= \left(2\ln\frac{\sqrt{3}}{2} + \frac{5}{4}\arctan\sqrt{3}\right) - \left(2\ln\frac{1}{2} + \frac{5}{4}\arctan 1\right)$

$= \left(\ln\frac{3}{4} + \frac{5}{4} \times \frac{\pi}{3}\right) - \left(\ln\frac{1}{4} + \frac{5}{4} \times \frac{\pi}{4}\right)$

$= \ln\frac{3}{4} - \ln\frac{1}{4} + \frac{5\pi}{12} - \frac{5\pi}{16}$

$= \frac{5\pi}{48} + \ln 3$

21)

$\int \frac{1+x}{1+x^2} \mathrm{d}x$

$= \int \left(\frac{1}{1+x^2} + \frac{x}{1+x^2}\right) \mathrm{d}x$

$= \arctan x + \frac{1}{2}\ln|1 + x^2| + c$

22) a)

$$\int \frac{2x+1}{1+4x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{2x}{1+4x^2} + \frac{1}{1+4x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{2x}{1+4x^2} + \frac{1}{1+(2x)^2}\right)\mathrm{d}x$$
$$= \frac{1}{4}\ln|1+4x^2| + \frac{1}{2}\arctan 2x + c$$

22) b)

$$\int \frac{3+5x}{4+x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{3}{4+x^2} + \frac{5x}{4+x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{3}{4\left(1+\frac{1}{4}x^2\right)} + \frac{5x}{4+x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{3}{4} \times \frac{1}{1+\left(\frac{1}{2}x\right)^2} + \frac{5x}{4+x^2}\right)\mathrm{d}x$$
$$= \frac{3}{4} \times \frac{1}{\frac{1}{2}}\arctan\left(\frac{1}{2}x\right) + \frac{5}{2}\ln|4+x^2| + c$$
$$= \frac{3}{2}\arctan\left(\frac{1}{2}x\right) + \frac{5}{2}\ln|4+x^2| + c$$

22) c)

$$\int \frac{2-3x}{9+4x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{2}{9+4x^2} - \frac{3x}{9+4x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{2}{9\left(1+\frac{4}{9}x^2\right)} - \frac{3x}{9+4x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{2}{9} \times \frac{1}{1+\left(\frac{2}{3}x\right)^2} - \frac{3x}{9+4x^2}\right)\mathrm{d}x$$
$$= \frac{2}{9} \times \frac{1}{\frac{2}{3}}\arctan\left(\frac{2}{3}x\right) - \frac{3}{8}\ln|9+4x^2| + c$$
$$= \frac{1}{3}\arctan\left(\frac{2}{3}x\right) - \frac{3}{8}\ln|9+4x^2| + c$$

22) d)

$$\int \frac{5-4x}{25+16x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{5}{25+16x^2} - \frac{4x}{25+16x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{5}{25\left(1+\frac{16}{25}x^2\right)} - \frac{4x}{25+16x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{1}{5} \times \frac{1}{1+\left(\frac{4}{5}x\right)^2} - \frac{4x}{25+16x^2}\right)\mathrm{d}x$$
$$= \frac{1}{5} \times \frac{1}{\frac{4}{5}} \arctan\left(\frac{4}{5}x\right) - \frac{1}{8}\ln|25+16x^2| + c$$
$$= \frac{1}{4}\arctan\left(\frac{4}{5}x\right) - \frac{1}{8}\ln|25+16x^2| + c$$

23)

$$\int \frac{1+x}{\sqrt{1-x^2}}\,\mathrm{d}x$$
$$= \int \left(\frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}\right)\mathrm{d}x$$
$$= \arcsin x - \sqrt{1-x^2} + c$$

24) a)

$$\int \frac{3x-1}{\sqrt{1-9x^2}}\,\mathrm{d}x$$
$$= \int \left(\frac{3x}{\sqrt{1-9x^2}} - \frac{1}{\sqrt{1-9x^2}}\right)\mathrm{d}x$$
$$= \int \left(\frac{3x}{\sqrt{1-9x^2}} - \frac{1}{\sqrt{1-(3x)^2}}\right)\mathrm{d}x$$
$$= -\frac{1}{3}\sqrt{1-9x^2} - \frac{1}{3}\arcsin 3x + c$$

24) b)

$$\int \frac{1+8x}{\sqrt{4-x^2}}\,\mathrm{d}x$$

$$= \int \left(\frac{1}{\sqrt{4-x^2}} + \frac{8x}{\sqrt{4-x^2}}\right)\mathrm{d}x$$

$$= \int \left(\frac{1}{\sqrt{4\left(1-\frac{1}{4}x^2\right)}} + \frac{8x}{\sqrt{4-x^2}}\right)\mathrm{d}x$$

$$= \int \left(\frac{1}{2} \times \frac{1}{\sqrt{1-\left(\frac{1}{2}x\right)^2}} + \frac{8x}{\sqrt{4-x^2}}\right)\mathrm{d}x$$

$$= \frac{1}{2} \times \frac{1}{\frac{1}{2}} \times \arcsin\left(\frac{1}{2}x\right) - 8\sqrt{4-x^2} + c$$

$$= \arcsin\left(\frac{1}{2}x\right) - 8\sqrt{4-x^2} + c$$

24) c)

$$\int \frac{6-x}{\sqrt{9-4x^2}}\,\mathrm{d}x$$

$$= \int \left(\frac{6}{\sqrt{9-4x^2}} - \frac{x}{\sqrt{9-4x^2}}\right)\mathrm{d}x$$

$$= \int \left(\frac{6}{\sqrt{9\left(1-\frac{4}{9}x^2\right)}} - \frac{x}{\sqrt{9-4x^2}}\right)\mathrm{d}x$$

$$= \int \left(2 \times \frac{1}{\sqrt{1-\left(\frac{2}{3}x\right)^2}} - \frac{x}{\sqrt{9-4x^2}}\right)\mathrm{d}x$$

$$= 2 \times \frac{1}{\frac{2}{3}} \times \arcsin\left(\frac{2}{3}x\right) + \frac{1}{4}\sqrt{9-4x^2} + c$$

$$= 3\arcsin\left(\frac{2}{3}x\right) + \frac{1}{4}\sqrt{9-4x^2} + c$$

24) d)

$$\int \frac{10-9x}{\sqrt{4-25x^2}}\,\mathrm{d}x$$

$$= \int \left(\frac{10}{\sqrt{4-25x^2}} - \frac{9x}{\sqrt{4-25x^2}}\right)\mathrm{d}x$$

$$= \int \left(\frac{10}{\sqrt{4\left(1-\frac{25}{4}x^2\right)}} - \frac{9x}{\sqrt{4-25x^2}}\right)\mathrm{d}x$$

$$= \int \left(5 \times \frac{1}{\sqrt{1-\left(\frac{5}{2}x\right)^2}} - \frac{9x}{\sqrt{4-25x^2}}\right)\mathrm{d}x$$

$$= 5 \times \frac{1}{\frac{5}{2}} \times \arcsin\left(\frac{5}{2}x\right) + \frac{9}{25}\sqrt{4-25x^2} + c$$

$$= 2\arcsin\left(\frac{5}{2}x\right) + \frac{9}{25}\sqrt{4-25x^2} + c$$

25)

$$\int_{-3}^{3} \frac{3+x}{9+x^2}\,\mathrm{d}x$$

$$= \int_{-3}^{3} \left(\frac{3}{9+x^2} + \frac{x}{9+x^2}\right)\mathrm{d}x$$

$$= \int_{-3}^{3} \left(\frac{3}{9\left(1+\frac{1}{9}x^2\right)} + \frac{x}{9+x^2}\right)\mathrm{d}x$$

$$= \int_{-3}^{3} \left(\frac{1}{3} \times \frac{1}{1+\left(\frac{1}{3}x\right)^2} + \frac{x}{9+x^2}\right)\mathrm{d}x$$

$$= \left[\frac{1}{3} \times \frac{1}{\frac{1}{3}} \times \arctan\left(\frac{1}{3}x\right) + \frac{1}{2}\ln|9+x^2|\right]_{-3}^{3}$$

$$= \left[\arctan\left(\frac{1}{3}x\right) + \frac{1}{2}\ln|9+x^2|\right]_{-3}^{3}$$

$$= \left(\frac{\pi}{4} + \frac{1}{2}\ln 18\right) - \left(-\frac{\pi}{4} + \frac{1}{2}\ln 18\right)$$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{2}\ \text{units}^2$$

26) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x+1}{\sqrt{9-x^2}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{\sqrt{9-x^2}} + \frac{1}{\sqrt{9-x^2}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{\sqrt{9-x^2}} + \frac{1}{\sqrt{9\left(1-\frac{1}{9}x^2\right)}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{\sqrt{9-x^2}} + \frac{1}{3} \times \frac{1}{\sqrt{1-\left(\frac{1}{3}x\right)^2}}$

$\Rightarrow y = -\sqrt{9-x^2} + \frac{1}{3} \times \frac{1}{\frac{1}{3}} \times \arcsin\frac{1}{3}x + c$

$\Rightarrow y = -\sqrt{9-x^2} + \arcsin\frac{1}{3}x + c$

26) b)

Let $x = 0$, $y = 3$:

$3 = -3 + \arcsin 0 + c \Rightarrow c = 6$

$\therefore y = -\sqrt{9-x^2} + \arcsin\frac{1}{3}x + 6$

27) a)

$\int \frac{x}{\sqrt{25-x^2}}\,\mathrm{d}x$

Use Integration by Substitution:

Let $u = 25 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x}\,\mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{25-x^2}}\,\mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x}\,\mathrm{d}u = -\frac{1}{2}\int u^{-\frac{1}{2}}\,\mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{25-x^2} + c$

27) b)

$\int \frac{1}{\sqrt{25-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{25\left(1-\frac{1}{25}x^2\right)}}\,\mathrm{d}x = \frac{1}{5}\int \frac{1}{\sqrt{1-\left(\frac{1}{5}x\right)^2}}\,\mathrm{d}x = \frac{1}{5} \times \frac{1}{\frac{1}{5}} \times \arcsin\left(\frac{1}{5}x\right) + c$

$= \arcsin\left(\frac{1}{5}x\right) + c$

27) c)

Mean Value:

$$= \frac{1}{0-\left(-\frac{5}{2}\right)} \int_{-\frac{5}{2}}^{0} \frac{5x+20}{\sqrt{25-x^2}}\,\mathrm{d}x$$

$$= \frac{2}{5} \int_{-\frac{5}{2}}^{0} \frac{5x+20}{\sqrt{25-x^2}}\,\mathrm{d}x$$

$$= 2 \int_{-\frac{5}{2}}^{0} \frac{x+4}{\sqrt{25-x^2}}\,\mathrm{d}x$$

$$= 2 \int_{-\frac{5}{2}}^{0} \left(\frac{x}{\sqrt{25-x^2}} + \frac{4}{\sqrt{25-x^2}}\right)\mathrm{d}x$$

$$= 2 \left[-\sqrt{25-x^2} + 4\arcsin\left(\frac{1}{5}x\right)\right]_{-\frac{5}{2}}^{0}$$

$$= 2\left(\left(-\sqrt{25-0^2} + 4\arcsin 0\right) - \left(-\sqrt{25-\left(-\frac{5}{2}\right)^2} + 4\arcsin\left(\frac{1}{5}\left(-\frac{5}{2}\right)\right)\right)\right)$$

$$= 2\left((-5+0) - \left(-\frac{5\sqrt{3}}{2} - \frac{2\pi}{3}\right)\right) = 2\left(-5 + \frac{5\sqrt{3}}{2} + \frac{2\pi}{3}\right) = -10 + 5\sqrt{3} + \frac{4\pi}{3}$$

28)

$$\int_{-\frac{\sqrt{2}}{3}}^{\frac{1}{3}} \frac{4-9x}{\sqrt{4-9x^2}}\,\mathrm{d}x$$

$$= \int_{-\frac{\sqrt{2}}{3}}^{\frac{1}{3}} \left(\frac{4}{\sqrt{4-9x^2}} - \frac{9x}{\sqrt{4-9x^2}}\right)\mathrm{d}x$$

$$= \int_{-\frac{\sqrt{2}}{3}}^{\frac{1}{3}} \left(\frac{4}{\sqrt{4\left(1-\frac{9}{4}x^2\right)}} - \frac{9x}{\sqrt{4-9x^2}}\right)\mathrm{d}x$$

$$= \int_{-\frac{\sqrt{2}}{3}}^{\frac{1}{3}} \left(2 \times \frac{1}{\sqrt{1-\left(\frac{3}{2}x\right)^2}} - \frac{9x}{\sqrt{4-9x^2}}\right)\mathrm{d}x$$

$$= \left[2 \times \frac{1}{\frac{3}{2}} \times \arcsin\left(\frac{3}{2}x\right) + \sqrt{4-9x^2}\right]_{-\frac{\sqrt{2}}{3}}^{\frac{1}{3}}$$

$$= \left[\frac{4}{3}\arcsin\left(\frac{3}{2}x\right) + \sqrt{4-9x^2}\right]_{-\frac{\sqrt{2}}{3}}^{\frac{1}{3}}$$

$$= \left(\frac{4}{3}\arcsin\left(\frac{3}{2} \times \frac{1}{3}\right) + \sqrt{4-9\left(\frac{1}{3}\right)^2}\right) - \left(\frac{4}{3}\arcsin\left(\frac{3}{2} \times -\frac{\sqrt{2}}{3}\right) + \sqrt{4-9\left(-\frac{\sqrt{2}}{3}\right)^2}\right)$$

$$= \left(\frac{2\pi}{9} + \sqrt{3}\right) - \left(-\frac{\pi}{3} + \sqrt{2}\right)$$

$= \left(\frac{5\pi}{9} + \sqrt{3} - \sqrt{2}\right)$ units^2

29) a)

Let $x = \sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \, \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{1-x^2}} \mathrm{d}x$

$= \int \frac{1}{\sqrt{1-\sin^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{\cos u}{\sqrt{\cos^2 u}} \mathrm{d}u$

$= \int \frac{\cos u}{\cos u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \arcsin x + c$

So $x = \sin u$ gives $\int \frac{1}{\sqrt{1-x^2}} \mathrm{d}x = \arcsin x + c$

29) b)

Let $x = \tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \, \mathrm{d}u$

$\therefore \int \frac{1}{1+x^2} \mathrm{d}x$

$= \int \frac{1}{1+\tan^2 u} \times \sec^2 u \, \mathrm{d}u$

$= \int \frac{\sec^2 u}{\sec^2 u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \arctan x + c$

So $x = \tan u$ gives $\int \frac{1}{1+x^2} \mathrm{d}x = \arctan x + c$

29) c)

Let $4x^2 = \sin^2 u \Rightarrow 2x = \sin u \Rightarrow x = \frac{1}{2}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cos u \Rightarrow \mathrm{d}x = \frac{1}{2}\cos u \,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{1-4x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{1-\sin^2 u}} \times \frac{1}{2}\cos u \,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arcsin 2x + c$

So $x = \frac{1}{2}\sin u$ gives $\int \frac{1}{\sqrt{1-4x^2}}\mathrm{d}x = \frac{1}{2}\arcsin 2x + c$

29) d)

Let $9x^2 = \tan^2 u \Rightarrow 3x = \tan u \Rightarrow x = \frac{1}{3}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\sec^2 u \Rightarrow \mathrm{d}x = \frac{1}{3}\sec^2 u \,\mathrm{d}u$

$\therefore \int \frac{1}{1+9x^2}\mathrm{d}x$

$= \int \frac{1}{1+\tan^2 u} \times \frac{1}{3}\sec^2 u \,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{3}\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\arctan 3x + c$

So $x = \frac{1}{3}\tan u$ gives $\int \frac{1}{1+9x^2}\mathrm{d}x = \frac{1}{3}\arctan 3x + c$

29) e)

Let $x^2 = 9\sin^2 u \Rightarrow x = 3\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\cos u \Rightarrow \mathrm{d}x = 3\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{9-x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9-9\sin^2 u}} \times 3\cos u\,\mathrm{d}u$

$= \int \frac{3\cos u}{\sqrt{9(1-\sin^2 u)}}\mathrm{d}u$

$= \int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \arcsin\left(\frac{1}{3}x\right) + c$

So $x = 3\sin u$ gives $\int \frac{1}{\sqrt{9-x^2}}\mathrm{d}x = \arcsin\left(\frac{1}{3}x\right) + c$

29) f)

Let $x^2 = 4\tan^2 u \Rightarrow x = 2\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sec^2 u \Rightarrow \mathrm{d}x = 2\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{4+x^2}\mathrm{d}x$

$= \int \frac{1}{4+4\tan^2 u} \times 2\sec^2 u\,\mathrm{d}u$

$= \int \frac{2\sec^2 u}{4(1+\tan^2 u)}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arctan\left(\frac{1}{2}x\right) + c$

So $x = 2\tan u$ gives $\int \frac{1}{4+x^2}\mathrm{d}x = \frac{1}{2}\arctan\left(\frac{1}{2}x\right) + c$

29) g)

Let $25x^2 = \sin^2 u \Rightarrow 5x = \sin u \Rightarrow x = \frac{1}{5}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{5}\cos u \Rightarrow \mathrm{d}x = \frac{1}{5}\cos u \,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{1-25x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{1-\sin^2 u}} \times \frac{1}{5}\cos u \,\mathrm{d}u$

$= \frac{1}{5}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{5}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{5}\mathrm{d}u$

$= \frac{1}{5}u + c$

$= \frac{1}{5}\arcsin 5x + c$

So $x = \frac{1}{5}\sin u$ gives $\int \frac{1}{\sqrt{1-25x^2}}\mathrm{d}x = \frac{1}{5}\arcsin 5x + c$

29) h)

Let $81x^2 = \tan^2 u \Rightarrow 9x = \tan u \Rightarrow x = \frac{1}{9}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{9}\sec^2 u \Rightarrow \mathrm{d}x = \frac{1}{9}\sec^2 u \,\mathrm{d}u$

$\therefore \int \frac{1}{1+81x^2}\mathrm{d}x$

$= \int \frac{1}{1+\tan^2 u} \times \frac{1}{9}\sec^2 u \,\mathrm{d}u$

$= \frac{1}{9}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{9}\mathrm{d}u$

$= \frac{1}{9}u + c$

$= \frac{1}{9}\arctan 9x + c$

So $x = \frac{1}{9}\tan u$ gives $\int \frac{1}{1+81x^2}\mathrm{d}x = \frac{1}{9}\arctan 9x + c$

29) i)

Let $9x^2 = 4\sin^2 u \Rightarrow 3x = 2\sin u \Rightarrow x = \frac{2}{3}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2}{3}\cos u \Rightarrow \mathrm{d}x = \frac{2}{3}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{4-9x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{4-4\sin^2 u}} \times \frac{2}{3}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{4(1-\sin^2 u)}} \times \frac{2}{3}\cos u\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{3}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{3}\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + c$

So $x = \frac{2}{3}\sin u$ gives $\int \frac{1}{\sqrt{4-9x^2}}\mathrm{d}x = \frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + c$

29) j)

Let $4x^2 = 9\tan^2 u \Rightarrow 2x = 3\tan u \Rightarrow x = \frac{3}{2}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\sec^2 u \Rightarrow \mathrm{d}x = \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{9+4x^2}\mathrm{d}x$

$= \int \frac{1}{9+9\tan^2 u} \times \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{9(1+\tan^2 u)} \times \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$= \frac{1}{6}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{6}\mathrm{d}u$

$= \frac{1}{6}u + c$

$= \frac{1}{6}\arctan\left(\frac{2x}{3}\right) + c$

So $x = \frac{3}{2}\tan u$ gives $\int \frac{1}{9+4x^2}\mathrm{d}x = \frac{1}{6}\arctan\left(\frac{2x}{3}\right) + c$

29) k)

Let $4x^2 = 25\sin^2 u \Rightarrow 2x = 5\sin u \Rightarrow x = \frac{5}{2}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5}{2}\cos u \Rightarrow \mathrm{d}x = \frac{5}{2}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{25-4x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{25-25\sin^2 u}} \times \frac{5}{2}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{25(1-\sin^2 u)}} \times \frac{5}{2}\cos u\,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arcsin\left(\frac{2x}{5}\right) + c$

So $x = \frac{5}{2}\sin u$ gives $\int \frac{1}{\sqrt{25-4x^2}}\mathrm{d}x = \frac{1}{2}\arcsin\left(\frac{2x}{5}\right) + c$

29) l)

Let $36x^2 = 49\tan^2 u \Rightarrow 6x = 7\tan u \Rightarrow x = \frac{7}{6}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{7}{6}\sec^2 u \Rightarrow \mathrm{d}x = \frac{7}{6}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{49+36x^2}\mathrm{d}x$

$= \int \frac{1}{49+49\tan^2 u} \times \frac{7}{6}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{49(1+\tan^2 u)} \times \frac{7}{6}\sec^2 u\,\mathrm{d}u$

$= \frac{1}{42}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{1}{42}\mathrm{d}u$

$= \frac{1}{42}u + c$

$= \frac{1}{42}\arctan\left(\frac{6x}{7}\right) + c$

So $x = \frac{7}{6}\tan u$ gives $\int \frac{1}{49+36x^2}\mathrm{d}x = \frac{1}{42}u + c$

29) m)

Let $3x^2 = 16\sin^2 u \Rightarrow \sqrt{3}x = 4\sin u \Rightarrow x = \frac{4}{\sqrt{3}}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{4\sqrt{3}}{3}\cos u \Rightarrow \mathrm{d}x = \frac{4\sqrt{3}}{3}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{16-3x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{16-16\sin^2 u}} \times \frac{4\sqrt{3}}{3}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{16(1-\sin^2 u)}} \times \frac{4\sqrt{3}}{3}\cos u\,\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{\sqrt{3}}{3}\mathrm{d}u$

$= \frac{\sqrt{3}}{3}u + c$

$= \frac{\sqrt{3}}{3}\arcsin\left(\frac{\sqrt{3}x}{4}\right) + c$

So $x = \frac{4}{\sqrt{3}}\sin u$ gives $\int \frac{1}{\sqrt{16-3x^2}}\mathrm{d}x = \frac{\sqrt{3}}{3}\arcsin\left(\frac{\sqrt{3}x}{4}\right) + c$

29) n)

Let $5x^2 = 4\tan^2 u \Rightarrow \sqrt{5}x = 2\tan u \Rightarrow x = \frac{2}{\sqrt{5}}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{5}}{5}\sec^2 u \Rightarrow \mathrm{d}x = \frac{2\sqrt{5}}{5}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{4+5x^2}\mathrm{d}x$

$= \int \frac{1}{4+4\tan^2 u} \times \frac{2\sqrt{5}}{5}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{4(1+\tan^2 u)} \times \frac{2\sqrt{5}}{5}\sec^2 u\,\mathrm{d}u$

$= \frac{\sqrt{5}}{10}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{5}}{10}\mathrm{d}u$

$= \frac{\sqrt{5}}{10}u + c$

$= \frac{\sqrt{5}}{10}\arctan\left(\frac{\sqrt{5}x}{2}\right) + c$

So $x = \frac{2}{\sqrt{5}}\tan u$ gives $\int \frac{1}{4+5x^2}\mathrm{d}x = \frac{\sqrt{5}}{10}\arctan\left(\frac{\sqrt{5}x}{2}\right) + c$

29) o)

Let $81x^2 = 8\sin^2 u \Rightarrow 9x = \sqrt{8}\sin u \Rightarrow x = \frac{\sqrt{8}}{9}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{2}}{9}\cos u \Rightarrow \mathrm{d}x = \frac{2\sqrt{2}}{9}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{8-81x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{8-8\sin^2 u}} \times \frac{2\sqrt{2}}{9}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{8(1-\sin^2 u)}} \times \frac{2\sqrt{2}}{9}\cos u\,\mathrm{d}u$

$= \frac{1}{9}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{9}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{9}\mathrm{d}u$

$= \frac{1}{9}u + c$

$= \frac{1}{9}\arcsin\left(\frac{9x}{\sqrt{8}}\right) + c$

$= \frac{1}{9}\arcsin\left(\frac{9\sqrt{2}x}{4}\right) + c$

So $x = \frac{\sqrt{8}}{9}\sin u$ gives $\int \frac{1}{\sqrt{8-81x^2}}\mathrm{d}x = \frac{1}{9}\arcsin\left(\frac{9\sqrt{2}x}{4}\right) + c$

29) p)

Let $25x^2 = 12\tan^2 u \Rightarrow 5x = \sqrt{12}\tan u \Rightarrow x = \frac{\sqrt{12}}{5}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{3}}{5}\sec^2 u \Rightarrow \mathrm{d}x = \frac{2\sqrt{3}}{5}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{12+25x^2}\mathrm{d}x$

$= \int \frac{1}{12+12\tan^2 u} \times \frac{2\sqrt{3}}{5}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{12(1+\tan^2 u)} \times \frac{2\sqrt{3}}{5}\sec^2 u\,\mathrm{d}u$

$= \frac{\sqrt{3}}{30}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{3}}{30}\mathrm{d}u$

$= \frac{\sqrt{3}}{30}u + c$

$= \frac{\sqrt{3}}{30}\arctan\left(\frac{5x}{\sqrt{12}}\right) + c$

$= \frac{\sqrt{3}}{30}\arctan\left(\frac{5\sqrt{3}x}{6}\right) + c$

So $x = \frac{\sqrt{12}}{5}\tan u$ gives $\int \frac{1}{12+25x^2}\mathrm{d}x = \frac{\sqrt{3}}{30}\arctan\left(\frac{5\sqrt{3}x}{6}\right) + c$

29) q)

Let $bx^2 = a\sin^2 u \Rightarrow \sqrt{b}x = \sqrt{a}\sin u \Rightarrow x = \frac{\sqrt{a}}{\sqrt{b}}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{a}}{\sqrt{b}}\cos u \Rightarrow \mathrm{d}x = \frac{\sqrt{a}}{\sqrt{b}}\cos u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{a-bx^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{a-a\sin^2 u}} \times \frac{\sqrt{a}}{\sqrt{b}}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{a(1-\sin^2 u)}} \times \frac{\sqrt{a}}{\sqrt{b}}\cos u\,\mathrm{d}u$

$= \frac{1}{\sqrt{b}}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{\sqrt{b}}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{\sqrt{b}}\mathrm{d}u$

$= \frac{1}{\sqrt{b}}u + c$

$= \frac{1}{\sqrt{b}}\arcsin\left(\frac{\sqrt{b}x}{\sqrt{a}}\right) + c$

$= \frac{1}{\sqrt{b}}\arcsin\left(\sqrt{\frac{b}{a}}x\right) + c$

So $x = \frac{\sqrt{a}}{\sqrt{b}}\sin u$ gives $\int \frac{1}{\sqrt{a-bx^2}}\mathrm{d}x = \frac{1}{\sqrt{b}}\arcsin\left(\sqrt{\frac{b}{a}}x\right) + c$

29) r)

Let $bx^2 = a\tan^2 u \Rightarrow \sqrt{b}x = \sqrt{a}\tan u \Rightarrow x = \frac{\sqrt{a}}{\sqrt{b}}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{a}}{\sqrt{b}}\sec^2 u \Rightarrow \mathrm{d}x = \frac{\sqrt{a}}{\sqrt{b}}\sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{a+bx^2}\mathrm{d}x$

$= \int \frac{1}{a+a\tan^2 u} \times \frac{\sqrt{a}}{\sqrt{b}}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{a(1+\tan^2 u)} \times \frac{\sqrt{a}}{\sqrt{b}}\sec^2 u\,\mathrm{d}u$

$= \frac{\sqrt{a}}{a\sqrt{b}}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{a}}{a\sqrt{b}}\mathrm{d}u$

$= \frac{\sqrt{a}}{a\sqrt{b}}u + c$

$= \frac{\sqrt{a}}{a\sqrt{b}}\arctan\left(\frac{\sqrt{b}x}{\sqrt{a}}\right) + c$

$= \frac{1}{\sqrt{ab}}\arctan\left(\sqrt{\frac{b}{a}}x\right) + c$

So $x = \frac{\sqrt{a}}{\sqrt{b}}\tan u$ gives $\int \frac{1}{a+bx^2}\mathrm{d}x = \frac{1}{\sqrt{ab}}\arctan\left(\sqrt{\frac{b}{a}}x\right) + c$

30) a)

Let $x^2 = \sin^2 u \Rightarrow x = \sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \, \mathrm{d}u$

$\therefore I = \int \sqrt{1 - x^2} \, \mathrm{d}x$

$= \int \sqrt{1 - \sin^2 u} \times \cos u \, \mathrm{d}u$

$= \int \sqrt{\cos^2 u} \times \cos u \, \mathrm{d}u$

$= \int \cos^2 u \, \mathrm{d}u$

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore I = \int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right) \mathrm{d}u$

$= \frac{1}{4}\sin 2u + \frac{1}{2}u + c$

$= \frac{1}{2}\sin u \cos u + \frac{1}{2}u + c$

$= \frac{1}{2}\sin u \sqrt{1 - \sin^2 u} + \frac{1}{2}u + c$

$= \frac{1}{2}x\sqrt{1 - x^2} + \frac{1}{2}\arcsin x + c$

So $x = \sin u$ gives $\int \sqrt{1 - x^2} \, \mathrm{d}x = \frac{1}{2}x\sqrt{1 - x^2} + \frac{1}{2}\arcsin x + c$

30) b)

Let $x^2 = \sin^2 u \Rightarrow x = \sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \, \mathrm{d}u$

$\therefore I = \int (1 - x^2)^{\frac{3}{2}} \mathrm{d}x$

$= \int (1 - \sin^2 u)^{\frac{3}{2}} \times \cos u \, \mathrm{d}u$

$= \int (\cos^2 u)^{\frac{3}{2}} \times \cos u \, \mathrm{d}u$

$= \int \cos^3 u \times \cos u \, \mathrm{d}u$

$= \int \cos^4 u \, \mathrm{d}u$

Now:

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \cos^4 u$

$= (\cos^2 u)^2$

$= \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cos^2 2u + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cos 4u + \frac{1}{2}\right) + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{8} + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$

$\therefore I = \int \left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right) \mathrm{d}u$

$= \frac{1}{32}\sin 4u + \frac{1}{4}\sin 2u + \frac{3}{8}u + c$

$= \frac{1}{32}(2\sin 2u \cos 2u) + \frac{1}{4}(2\sin u \cos u) + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sin u \cos u)(1 - 2\sin^2 u) + \frac{1}{2}\sin u \cos u + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sin u \cos u - 4\sin^3 u \cos u) + \frac{1}{2}\sin u \cos u + \frac{3}{8}u + c$

$= \frac{1}{8}\sin u \cos u - \frac{1}{4}\sin^3 u \cos u + \frac{1}{2}\sin u \cos u + \frac{3}{8}u + c$

$= \frac{5}{8}\sin u \cos u - \frac{1}{4}\sin^3 u \cos u + \frac{3}{8}u + c$

$= \frac{5}{8}\sin u \sqrt{1 - \sin^2 u} - \frac{1}{4}\sin^3 u \sqrt{1 - \sin^2 u} + \frac{3}{8}u + c$

$= \frac{5}{8}x\sqrt{1 - x^2} - \frac{1}{4}x^3\sqrt{1 - x^2} + \frac{3}{8}\arcsin x + c$

$= \frac{1}{8}x(5 - 2x^2)\sqrt{1 - x^2} + \frac{3}{8}\arcsin x + c$

So $x = \sin u$ gives $\int (1 - x^2)^{\frac{3}{2}} \mathrm{d}x = \frac{1}{8}x(5 - 2x^2)\sqrt{1 - x^2} + \frac{3}{8}\arcsin x + c$

31) a)

Let $x^2 = \tan^2 u \Rightarrow x = \tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \, \mathrm{d}u$

$\therefore I = \int \frac{1}{(1+x^2)^2} \mathrm{d}x$

$= \int \frac{1}{(1+\tan^2 u)^2} \times \sec^2 u \, \mathrm{d}u$

$= \int \frac{\sec^2 u}{(\sec^2 u)^2} \mathrm{d}u$

$= \int \frac{1}{\sec^2 u} \mathrm{d}u$

$= \int \cos^2 u \, \mathrm{d}u$

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore I = \int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right) \mathrm{d}u$

$= \frac{1}{4}\sin 2u + \frac{1}{2}u + c$

$= \frac{1}{4}(2\sin u \cos u) + \frac{1}{2}u + c$

$= \frac{1}{2}\sin u \cos u + \frac{1}{2}u + c$

If $\tan u = \frac{x}{1}$ then $\sin u = \frac{x}{\sqrt{1+x^2}}$ and $\cos u = \frac{1}{\sqrt{1+x^2}}$

$\therefore I = \frac{1}{2} \times \frac{x}{\sqrt{1+x^2}} \times \frac{1}{\sqrt{1+x^2}} + \frac{1}{2}\arctan x + c$

$= \frac{x}{2(1+x^2)} + \frac{1}{2}\arctan x + c$

So $x = \tan u$ gives $\int \frac{1}{(1+x^2)^2} \mathrm{d}x = \frac{x}{2(1+x^2)} + \frac{1}{2}\arctan x + c$

31) b)

Let $x^2 = 4\tan^2 u \Rightarrow x = 2\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sec^2 u \Rightarrow \mathrm{d}x = 2\sec^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{1}{(4+x^2)^2}\,\mathrm{d}x$

$= \int \frac{1}{(4+4\tan^2 u)^2} \times 2\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{\left(4(1+\tan^2 u)\right)^2} \times 2\sec^2 u\,\mathrm{d}u$

$= \frac{1}{8}\int \frac{\sec^2 u}{(\sec^2 u)^2}\,\mathrm{d}u$

$= \frac{1}{8}\int \frac{1}{\sec^2 u}\,\mathrm{d}u$

$= \frac{1}{8}\int \cos^2 u\,\mathrm{d}u$

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore I = \frac{1}{8}\int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{1}{16}\cos 2u + \frac{1}{16}\right)\mathrm{d}u$

$= \frac{1}{32}\sin 2u + \frac{1}{16}u + c$

$= \frac{1}{32}(2\sin u\cos u) + \frac{1}{16}u + c$

$= \frac{1}{16}\sin u\cos u + \frac{1}{16}u + c$

If $\tan u = \frac{x}{2}$ then $\sin u = \frac{x}{\sqrt{4+x^2}}$ and $\cos u = \frac{2}{\sqrt{4+x^2}}$

$\therefore I = \frac{1}{16} \times \frac{x}{\sqrt{4+x^2}} \times \frac{2}{\sqrt{4+x^2}} + \frac{1}{16}\arctan\left(\frac{1}{2}x\right) + c$

$= \frac{x}{8(4+x^2)} + \frac{1}{16}\arctan\left(\frac{1}{2}x\right) + c$

So $x = 2\tan u$ gives $\int \frac{1}{(4+x^2)^2}\,\mathrm{d}x = \frac{x}{8(4+x^2)} + \frac{1}{16}\arctan\left(\frac{1}{2}x\right) + c$

31) c)

Let $x^2 = \tan^2 u \Rightarrow x = \tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \, \mathrm{d}u$

$\therefore I = \int \frac{1}{(1+x^2)^3} \mathrm{d}x$

$= \int \frac{1}{(1+\tan^2 u)^3} \times \sec^2 u \, \mathrm{d}u$

$= \int \frac{\sec^2 u}{(\sec^2 u)^3} \mathrm{d}u$

$= \int \frac{1}{(\sec^2 u)^2} \mathrm{d}u$

$= \int \frac{1}{\sec^4 u} \mathrm{d}u$

$= \int \cos^4 u \, \mathrm{d}u$

Now:

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \cos^4 u$

$= (\cos^2 u)^2$

$= \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cos^2 2u + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cos 4u + \frac{1}{2}\right) + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{8} + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$

$\therefore I = \int \left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right) \mathrm{d}u$

$= \frac{1}{32}\sin 4u + \frac{1}{4}\sin 2u + \frac{3}{8}u + c$

$= \frac{1}{32}(2\sin 2u \cos 2u) + \frac{1}{4}(2\sin u \cos u) + \frac{3}{8}u + c$

$= \frac{1}{16}\sin 2u \cos 2u + \frac{1}{2}\sin u \cos u + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sin u \cos u)(2\cos^2 u - 1) + \frac{1}{2}\sin u \cos u + \frac{3}{8}u + c$

$= \frac{1}{4}\sin u \cos^3 u - \frac{1}{8}\sin u \cos u + \frac{1}{2}\sin u \cos u + \frac{3}{8}u + c$

$= \frac{1}{4}\sin u \cos^3 u + \frac{3}{8}\sin u \cos u + \frac{3}{8}u + c$

If $\tan u = \frac{x}{1}$ then $\sin u = \frac{x}{\sqrt{1+x^2}}$ and $\cos u = \frac{1}{\sqrt{1+x^2}}$

$\therefore I = \frac{1}{4} \times \frac{x}{\sqrt{1+x^2}} \times \left(\frac{1}{\sqrt{1+x^2}}\right)^3 + \frac{3}{8} \times \frac{x}{\sqrt{1+x^2}} \times \frac{1}{\sqrt{1+x^2}} + \frac{3}{8}\arctan x + c$

$= \frac{x}{4(1+x^2)^2} + \frac{3x}{8(1+x^2)} + \frac{3}{8}\arctan x + c$

$= \frac{2x}{8(1+x^2)^2} + \frac{3x(1+x^2)}{8(1+x^2)^2} + \frac{3}{8}\arctan x + c$

$= \frac{2x+3x(1+x^2)}{8(1+x^2)^2} + \frac{3}{8}\arctan x + c$

$= \frac{5x+3x^3}{8(1+x^2)^2} + \frac{3}{8}\arctan x + c$

So $x = \tan u$ gives $\int \frac{1}{(1+x^2)^3}\,\mathrm{d}x = \frac{5x+3x^3}{8(1+x^2)^2} + \frac{3}{8}\arctan x + c$

31) d)

Let $4x^2 = 9\tan^2 u \Rightarrow 2x = 3\tan u \Rightarrow x = \frac{3}{2}\tan u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\sec^2 u \Rightarrow \mathrm{d}x = \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{1}{(9+4x^2)^3}\,\mathrm{d}x$

$= \int \frac{1}{(9+9\tan^2 u)^3} \times \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{\left(9(1+\tan^2 u)\right)^3} \times \frac{3}{2}\sec^2 u\,\mathrm{d}u$

$= \frac{1}{486}\int \frac{\sec^2 u}{(1+\tan^2 u)^3}\,\mathrm{d}u$

$= \frac{1}{486}\int \frac{\sec^2 u}{(\sec^2 u)^3}\,\mathrm{d}u$

$= \frac{1}{486}\int \frac{1}{(\sec^2 u)^2}\,\mathrm{d}u$

$= \frac{1}{486}\int \frac{1}{\sec^4 u}\,\mathrm{d}u$

$= \frac{1}{486}\int \cos^4 u\,\mathrm{d}u$

Now:

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \cos^4 u$

$= (\cos^2 u)^2$

$= \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cos^2 2u + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cos 4u + \frac{1}{2}\right) + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{8} + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$

$\therefore I = \frac{1}{486}\int\left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int\left(\frac{1}{3888}\cos 4u + \frac{1}{972}\cos 2u + \frac{1}{1296}\right)\mathrm{d}u$

$= \frac{1}{15552}\sin 4u + \frac{1}{1944}\sin 2u + \frac{1}{1296}u + c$

$= \frac{1}{15552}(2\sin 2u\cos 2u) + \frac{1}{1944}(2\sin u\cos u) + \frac{1}{1296}u + c$

$= \frac{1}{7776}\sin 2u\cos 2u + \frac{1}{972}\sin u\cos u + \frac{1}{1296}u + c$

$= \frac{1}{7776}(2\sin u\cos u)(2\cos^2 u - 1) + \frac{1}{972}\sin u\cos u + \frac{1}{1296}u + c$

$= \frac{1}{1944}\sin u\cos^3 u - \frac{1}{3888}\sin u\cos u + \frac{1}{972}\sin u\cos u + \frac{1}{1296}u + c$

$= \frac{1}{1944}\sin u\cos^3 u + \frac{1}{1296}\sin u\cos u + \frac{1}{1296}u + c$

If $\tan u = \frac{2x}{3}$ then $\sin u = \frac{2x}{\sqrt{9+4x^2}}$ and $\cos u = \frac{3}{\sqrt{9+4x^2}}$

$\therefore I = \frac{1}{1944}\times\frac{2x}{\sqrt{9+4x^2}}\times\left(\frac{3}{\sqrt{9+4x^2}}\right)^3 + \frac{1}{1296}\times\frac{2x}{\sqrt{9+4x^2}}\times\frac{3}{\sqrt{9+4x^2}} + \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + c$

$= \frac{x}{36(9+4x^2)^2} + \frac{x}{216(9+4x^2)} + \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + c$

$= \frac{6x}{216(9+4x^2)^2} + \frac{x(9+4x^2)}{216(9+4x^2)^2} + \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + c$

$= \frac{6x+x(9+4x^2)}{216(9+4x^2)^2} + \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + c$

$= \frac{15x+4x^3}{216(9+4x^2)^2} + \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + c$

So $x = \frac{3}{2}\tan u$ gives $\int\frac{1}{(9+4x^2)^3}\mathrm{d}x = \frac{15x+4x^3}{216(9+4x^2)^2} + \frac{1}{1296}\arctan\left(\frac{2x}{3}\right) + c$

32) a)

Let $x^2 = 36\sin^2 u \Rightarrow x = 6\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 6\cos u \Rightarrow \mathrm{d}x = 6\cos u\,\mathrm{d}u$

$\therefore I = \int \sqrt{36 - x^2}\,\mathrm{d}x$

$= \int \sqrt{36 - 36\sin^2 u} \times 6\cos u\,\mathrm{d}u$

$= \int \sqrt{36(1 - \sin^2 u)} \times 6\cos u\,\mathrm{d}u$

$= 36\int \sqrt{1 - \sin^2 u} \times \cos u\,\mathrm{d}u$

$= 36\int \sqrt{\cos^2 u} \times \cos u\,\mathrm{d}u$

$= 36\int \cos^2 u\,\mathrm{d}u$

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore I = 36\int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int (18\cos 2u + 18)\,\mathrm{d}u$

$= 9\sin 2u + 18u + c$

$= 9(2\sin u\cos u) + 18u + c$

$= 18\sin u\cos u + 18u + c$

$= 18\sin u\sqrt{1 - \sin^2 u} + 18u + c$

$= 18\left(\frac{1}{6}x\right)\sqrt{1 - \left(\frac{1}{6}x\right)^2} + 18\arcsin\left(\frac{1}{6}x\right) + c$

$= 3x\sqrt{1 - \frac{1}{36}x^2} + 18\arcsin\left(\frac{1}{6}x\right) + c$

$= 3x\sqrt{\frac{1}{36}(36 - x^2)} + 18\arcsin\left(\frac{1}{6}x\right) + c$

$= \frac{1}{2}x\sqrt{36 - x^2} + 18\arcsin\left(\frac{1}{6}x\right) + c$

So $x = 6\sin u$ gives $\int \sqrt{36 - x^2}\,\mathrm{d}x = \frac{1}{2}x\sqrt{36 - x^2} + 18\arcsin\left(\frac{1}{6}x\right) + c$

32) b)

Let $4x^2 = \sin^2 u \Rightarrow 2x = \sin u \Rightarrow x = \frac{1}{2}\sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cos u \Rightarrow \mathrm{d}x = \frac{1}{2}\cos u\,\mathrm{d}u$

$\therefore I = \int (1 - 4x^2)^{\frac{3}{2}}\,\mathrm{d}x$

$= \int (1 - \sin^2 u)^{\frac{3}{2}} \times \frac{1}{2}\cos u\,\mathrm{d}u$

$= \frac{1}{2}\int (\cos^2 u)^{\frac{3}{2}} \times \cos u\,\mathrm{d}u$

$= \frac{1}{2}\int \cos^3 u \times \cos u \, du$

$= \frac{1}{2}\int \cos^4 u \, du$

Now:

$\cos 2A \equiv 2\cos^2 A - 1 \Leftrightarrow \cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \cos^4 u$

$= (\cos^2 u)^2$

$= \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cos^2 2u + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cos 4u + \frac{1}{2}\right) + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{8} + \frac{1}{2}\cos 2u + \frac{1}{4}$

$= \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$

$\therefore I = \frac{1}{2}\int \left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right) du$

$= \frac{1}{2}\left(\frac{1}{32}\sin 4u + \frac{1}{4}\sin 2u + \frac{3}{8}u\right) + c$

$= \frac{1}{64}\sin 4u + \frac{1}{8}\sin 2u + \frac{3}{16}u + c$

$= \frac{1}{64}(2\sin 2u \cos 2u) + \frac{1}{8}(2\sin u \cos u) + \frac{3}{16}u + c$

$= \frac{1}{32}(2\sin u \cos u)(1 - 2\sin^2 u) + \frac{1}{4}\sin u \cos u + \frac{3}{16}u + c$

$= \frac{1}{32}(2\sin u \cos u - 4\sin^3 u \cos u) + \frac{1}{4}\sin u \cos u + \frac{3}{16}u + c$

$= \frac{1}{16}\sin u \cos u - \frac{1}{8}\sin^3 u \cos u + \frac{1}{4}\sin u \cos u + \frac{3}{16}u + c$

$= \frac{5}{16}\sin u \cos u - \frac{1}{8}\sin^3 u \cos u + \frac{3}{16}u + c$

$= \frac{5}{16}\sin u \sqrt{1 - \sin^2 u} - \frac{1}{8}\sin^3 u \sqrt{1 - \sin^2 u} + \frac{3}{16}u + c$

$= \frac{5}{16}(2x)\sqrt{1 - (2x)^2} - \frac{1}{8}(2x)^3\sqrt{1 - (2x)^2} + \frac{3}{16}\arcsin 2x + c$

$= \frac{5}{8}x\sqrt{1 - 4x^2} - x^3\sqrt{1 - 4x^2} + \frac{3}{16}\arcsin 2x + c$

$= \frac{1}{8}x(5 - 8x^2)\sqrt{1 - 4x^2} + \frac{3}{16}\arcsin 2x + c$

So $x = \frac{1}{2}\sin u$ gives $\int (1 - 4x^2)^{\frac{3}{2}} dx = \frac{1}{8}x(5 - 8x^2)\sqrt{1 - 4x^2} + \frac{3}{16}\arcsin 2x + c$

33) a)

$26 - 10x + x^2$

$\equiv 26 + x^2 - 10x$

$\equiv 26 + (x - 5)^2 - 25$

$\equiv 1 + (x - 5)^2$

33) b)

$101 + 20x + x^2$

$\equiv 101 + x^2 + 20x$

$\equiv 101 + (x + 10)^2 - 100$

$\equiv 1 + (x + 10)^2$

33) c)

$2 + 4x + 4x^2$

$\equiv 2 + 4x^2 + 4x$

$\equiv 2 + 4(x^2 + x)$

$\equiv 2 + 4\left(\left(x + \frac{1}{2}\right)^2 - \frac{1}{4}\right)$

$\equiv 2 + 4\left(x + \frac{1}{2}\right)^2 - 1$

$\equiv 1 + 4\left(x + \frac{1}{2}\right)^2$

33) d)

$5 - 12x + 9x^2$

$\equiv 5 + 9x^2 - 12x$

$\equiv 5 + 9\left(x^2 - \frac{4}{3}x\right)$

$\equiv 5 + 9\left(\left(x - \frac{2}{3}\right)^2 - \frac{4}{9}\right)$

$\equiv 5 + 9\left(x - \frac{2}{3}\right)^2 - 4$

$\equiv 1 + 9\left(x - \frac{2}{3}\right)^2$

33) e)

$25 - 8x + x^2$
$\equiv 25 + x^2 - 8x$
$\equiv 25 + (x-4)^2 - 16$
$\equiv 9 + (x-4)^2$

33) f)

$29 + 8x + 4x^2$
$\equiv 29 + 4x^2 + 8x$
$\equiv 29 + 4(x^2 + 2x)$
$\equiv 29 + 4((x+1)^2 - 1)$
$\equiv 29 + 4(x+1)^2 - 4$
$\equiv 25 + 4(x+1)^2$

34) a)

$2 + 2x + x^2$
$\equiv 2 + x^2 + 2x$
$\equiv 2 + (x+1)^2 - 1$
$\equiv 1 + (x+1)^2$

34) b)

Let $x + 1 = \tan u \Rightarrow x = \tan u - 1 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \ \mathrm{d}u$
$\therefore \int \frac{1}{2+2x+x^2}\,\mathrm{d}x = \int \frac{1}{1+(x+1)^2}\,\mathrm{d}x = \int \frac{1}{1+\tan^2 u} \times \sec^2 u\,\mathrm{d}u$
$= \int \frac{\sec^2 u}{\sec^2 u}\,\mathrm{d}u$
$= \int 1\,\mathrm{d}u$
$= u + c$
$= \arctan(x+1) + c$

35) a)

$\int \frac{1}{10-6x+x^2}\,\mathrm{d}x$
$= \int \frac{1}{10+(x-3)^2-9}\,\mathrm{d}x$
$= \int \frac{1}{1+(x-3)^2}\,\mathrm{d}x$
$= \arctan(x-3) + c$

35) b)

$\int \frac{1}{65+16x+x^2}\,\mathrm{d}x$

$= \int \frac{1}{65+(x+8)^2-64}\,\mathrm{d}x$

$= \int \frac{1}{1+(x+8)^2}\,\mathrm{d}x$

$= \arctan(x+8) + c$

36) a)

$10 + 18x + 9x^2$

$\equiv 10 + 9x^2 + 18x$

$\equiv 10 + 9(x^2 + 2x)$

$\equiv 10 + 9((x+1)^2 - 1)$

$\equiv 10 + 9(x+1)^2 - 9$

$\equiv 1 + 9(x+1)^2$

36) b)

Let $3x + 3 = \tan u \Rightarrow x = \frac{1}{3}\tan u - 1 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\sec^2 u \Rightarrow \mathrm{d}x = \frac{1}{3}\sec^2 u\ \mathrm{d}u$

$\therefore \int \frac{1}{10+18x+9x^2}\,\mathrm{d}x = \int \frac{1}{1+9(x+1)^2}\,\mathrm{d}x = \int \frac{1}{1+\left(3(x+1)\right)^2}\,\mathrm{d}x = \int \frac{1}{1+(3x+3)^2}\,\mathrm{d}x$

$= \int \frac{1}{1+\tan^2 u} \times \frac{1}{3}\sec^2 u\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\sec^2 u}{\sec^2 u}\,\mathrm{d}u$

$= \frac{1}{3}\int 1\,\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\arctan(3x+3) + c$

37) a)

$$\int \frac{1}{10+12x+4x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{10+4(x^2+3x)}\,\mathrm{d}x$$
$$= \int \frac{1}{10+4\left(\left(x+\frac{3}{2}\right)^2-\frac{9}{4}\right)}\,\mathrm{d}x$$
$$= \int \frac{1}{10+4\left(x+\frac{3}{2}\right)^2-9}\,\mathrm{d}x$$
$$= \int \frac{1}{1+(2x+3)^2}\,\mathrm{d}x$$
$$= \frac{1}{2}\arctan(2x+3)+c$$

37) b)

$$\int \frac{1}{50-28x+4x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{50+4(x^2-7x)}\,\mathrm{d}x$$
$$= \int \frac{1}{50+4\left(\left(x-\frac{7}{2}\right)^2-\frac{49}{4}\right)}\,\mathrm{d}x$$
$$= \int \frac{1}{50+4\left(x-\frac{7}{2}\right)^2-49}\,\mathrm{d}x$$
$$= \int \frac{1}{1+(2x-7)^2}\,\mathrm{d}x$$
$$= \frac{1}{2}\arctan(2x-7)+c$$

38) a)

$$13-6x+x^2$$
$$\equiv 13+x^2-6x$$
$$\equiv 13+(x-3)^2-9$$
$$\equiv 4+(x-3)^2$$

38) b)

Let $\frac{x-3}{2} = \tan u \Rightarrow x = 2\tan u + 3 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sec^2 u \Rightarrow \mathrm{d}x = 2\sec^2 u \ \mathrm{d}u$

$\therefore \int \frac{1}{13-6x+x^2}\mathrm{d}x = \int \frac{1}{4+(x-3)^2}\mathrm{d}x = \int \frac{1}{4\left(1+\frac{1}{4}(x-3)^2\right)}\mathrm{d}x = \frac{1}{4}\int \frac{1}{1+\left(\frac{x-3}{2}\right)^2}\mathrm{d}x$

$= \frac{1}{4}\int \frac{1}{1+\tan^2 u} \times 2\sec^2 u \, \mathrm{d}u$

$= \frac{1}{2}\int \frac{\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \frac{1}{2}\int 1 \, \mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arctan\left(\frac{x-3}{2}\right) + c$

39) a)

$\int \frac{1}{26+8x+2x^2}\mathrm{d}x$

$= \int \frac{1}{26+2(x^2+4x)}\mathrm{d}x$

$= \int \frac{1}{26+2((x+2)^2-4)}\mathrm{d}x$

$= \int \frac{1}{26+2(x+2)^2-8}\mathrm{d}x$

$= \int \frac{1}{18+2(x+2)^2}\mathrm{d}x$

$= \int \frac{1}{18\left(1+\frac{1}{9}(x+2)^2\right)}\mathrm{d}x$

$= \frac{1}{18}\int \frac{1}{1+\left(\frac{x+2}{3}\right)^2}\mathrm{d}x$

$= \frac{1}{18} \times \frac{1}{\frac{1}{3}} \times \arctan\left(\frac{x+2}{3}\right) + c$

$= \frac{1}{6}\arctan\left(\frac{x+2}{3}\right) + c$

39) b)

$$\int \frac{6}{5+12x+8x^2}\,\mathrm{d}x$$
$$= \int \frac{6}{5+8\left(x^2+\frac{3}{2}x\right)}\,\mathrm{d}x$$
$$= \int \frac{6}{5+8\left(\left(x+\frac{3}{4}\right)^2-\frac{9}{16}\right)}\,\mathrm{d}x$$
$$= \int \frac{6}{5+8\left(x+\frac{3}{4}\right)^2-\frac{9}{2}}\,\mathrm{d}x$$
$$= \int \frac{6}{\frac{1}{2}+8\left(x+\frac{3}{4}\right)^2}\,\mathrm{d}x$$
$$= \int \frac{6}{\frac{1}{2}\left(1+16\left(x+\frac{3}{4}\right)^2\right)}\,\mathrm{d}x$$
$$= 12\int \frac{1}{1+(4x+3)^2}\,\mathrm{d}x$$
$$= 12 \times \frac{1}{4} \times \arctan(4x+3) + c$$
$$= 3\arctan(4x+3) + c$$

40)

$$\int_{\frac{7}{3}}^{3} \frac{2}{65-48x+9x^2}\,\mathrm{d}x$$
$$= \int_{\frac{7}{3}}^{3} \frac{2}{65+9\left(x^2-\frac{16}{3}x\right)}\,\mathrm{d}x$$
$$= \int_{\frac{7}{3}}^{3} \frac{2}{65+9\left(\left(x-\frac{8}{3}\right)^2-\frac{64}{9}\right)}\,\mathrm{d}x$$
$$= \int_{\frac{7}{3}}^{3} \frac{2}{65+9\left(x-\frac{8}{3}\right)^2-64}\,\mathrm{d}x$$
$$= \int_{\frac{7}{3}}^{3} \frac{2}{1+9\left(x-\frac{8}{3}\right)^2}\,\mathrm{d}x$$
$$= 2\int_{\frac{7}{3}}^{3} \frac{1}{1+(3x-8)^2}\,\mathrm{d}x$$
$$= \left[\frac{2}{3}\arctan(3x-8)\right]_{\frac{7}{3}}^{3}$$
$$= \left(\frac{2}{3}\arctan(3\times 3-8)\right) - \left(\frac{2}{3}\arctan\left(3\times\frac{7}{3}-8\right)\right)$$
$$= \left(\frac{2}{3}\arctan 1\right) - \left(\frac{2}{3}\arctan(-1)\right)$$
$$= \left(\frac{\pi}{6}\right) - \left(-\frac{\pi}{6}\right)$$
$$= \frac{\pi}{3}\ \text{units}^2$$

41)

$$\int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{7}{41+10x+x^2}\,\mathrm{d}x$$
$$= \int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{7}{41+(x+5)^2-25}\,\mathrm{d}x$$
$$= \int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{7}{16+(x+5)^2}\,\mathrm{d}x$$
$$= \frac{7}{16}\int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{1}{1+\frac{1}{16}(x+5)^2}\,\mathrm{d}x$$
$$= \frac{7}{16}\int_{-4\sqrt{3}-5}^{4\sqrt{3}-5} \frac{1}{1+\left(\frac{x+5}{4}\right)^2}\,\mathrm{d}x$$
$$= \left[\frac{7}{16}\times\frac{1}{\frac{1}{4}}\times\arctan\left(\frac{x+5}{4}\right)\right]_{-4\sqrt{3}-5}^{4\sqrt{3}-5}$$
$$= \left[\frac{7}{4}\arctan\left(\frac{x+5}{4}\right)\right]_{-4\sqrt{3}-5}^{4\sqrt{3}-5}$$
$$= \left(\frac{7}{4}\arctan\left(\frac{(4\sqrt{3}-5)+5}{4}\right)\right) - \left(\frac{7}{4}\arctan\left(\frac{(-4\sqrt{3}-5)+5}{4}\right)\right)$$
$$= \left(\frac{7}{4}\arctan\left(\sqrt{3}\right)\right) - \left(\frac{7}{4}\arctan\left(-\sqrt{3}\right)\right)$$
$$= \left(\frac{7\pi}{12}\right) - \left(-\frac{7\pi}{12}\right)$$
$$= \frac{7\pi}{6}\ \text{units}^2$$

42) a)

$$-8+6x-x^2$$
$$\equiv -8-(x^2-6x)$$
$$\equiv -8-((x-3)^2-9)$$
$$\equiv -8-(x-3)^2+9$$
$$\equiv 1-(x-3)^2$$

42) b)

$$-48-14x-x^2$$
$$\equiv -48-(x^2+14x)$$
$$\equiv -48-((x+7)^2-49)$$
$$\equiv -48-(x+7)^2+49$$
$$\equiv 1-(x+7)^2$$

42) c)

$-15 + 16x - 4x^2$
$\equiv -15 - 4(x^2 - 4x)$
$\equiv -15 - 4((x-2)^2 - 4)$
$\equiv -15 - 4(x-2)^2 + 16$
$\equiv 1 - 4(x-2)^2$

42) d)

$-224 - 90x - 9x^2$
$\equiv -224 - 9(x^2 + 10x)$
$\equiv -224 - 9((x+5)^2 - 25)$
$\equiv -224 - 9(x+5)^2 + 225$
$\equiv 1 - 9(x+5)^2$

42) e)

$-32 + 12x - x^2$
$\equiv -32 - (x^2 - 12x)$
$\equiv -32 - ((x-6)^2 - 36)$
$\equiv -32 - (x-6)^2 + 36$
$\equiv 4 - (x-6)^2$

42) f)

$-128 - 48x - 4x^2$
$\equiv -128 - 4(x^2 + 12x)$
$\equiv -128 - 4((x+6)^2 - 36)$
$\equiv -128 - 4(x+6)^2 + 144$
$\equiv 16 - 4(x+6)^2$

43) a)

$-63 + 16x - x^2$
$\equiv -63 - (x^2 - 16x)$
$\equiv -63 - ((x-8)^2 - 64)$
$\equiv -63 - (x-8)^2 + 64$
$\equiv 1 - (x-8)^2$

43) b)

Let $x-8=\sin u \Rightarrow x=\sin u+8 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=\cos u \Rightarrow \mathrm{d}x=\cos u\ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{-63+16x-x^2}}\mathrm{d}x=\int \frac{1}{\sqrt{1-(x-8)^2}}\mathrm{d}x=\int \frac{1}{\sqrt{1-\sin^2 u}}\times \cos u\,\mathrm{d}u=\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}x$

$=\int \frac{\cos u}{\cos u}\mathrm{d}u$

$=\int 1\,\mathrm{d}u$

$=u+c$

$=\arcsin(x-8)+c$

44) a)

$\int \frac{1}{\sqrt{-3-4x-x^2}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-3-(x^2+4x)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-3-((x+2)^2-4)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-3-(x+2)^2+4}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{1-(x+2)^2}}\mathrm{d}x$

$=\arcsin(x+2)+c$

44) b)

$\int \frac{1}{\sqrt{-80+18x-x^2}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-80-(x^2-18x)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-80-((x-9)^2-81)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-80-(x-9)^2+81}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{1-(x-9)^2}}\mathrm{d}x$

$=\arcsin(x-9)+c$

45) a)

$-35-24x-4x^2$

$\equiv -35-4(x^2+6x)$

$\equiv -35-4((x+3)^2-9)$

$\equiv -35-4(x+3)^2+36$

$\equiv 1-4(x+3)^2$

45) b)

Let $2x+6=\sin u \Rightarrow x=\frac{1}{2}\sin u-3 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=\frac{1}{2}\cos u \Rightarrow \mathrm{d}x=\frac{1}{2}\cos u\ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{-35-24x-4x^2}}\mathrm{d}x=\int \frac{1}{\sqrt{1-4(x+3)^2}}\mathrm{d}x=\int \frac{1}{\sqrt{1-(2x+6)^2}}\mathrm{d}x=\int \frac{1}{\sqrt{1-\sin^2 u}}\times\frac{1}{2}\cos u\,\mathrm{d}u$

$=\frac{1}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}x$

$=\frac{1}{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$=\int \frac{1}{2}\mathrm{d}u$

$=\frac{1}{2}u+c$

$=\frac{1}{2}\arcsin(2x+6)+c$

46) a)

$\int \frac{1}{\sqrt{-63-32x-4x^2}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-63-4(x^2+8x)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-63-4((x+4)^2-16)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-63-4(x+4)^2+64}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{1-4(x+4)^2}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{1-(2x+8)^2}}\mathrm{d}x$

$=\frac{1}{2}\arcsin(2x+8)+c$

46) b)

$\int \frac{1}{\sqrt{-99+100x-25x^2}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-99-25(x^2-4x)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-99-25((x-2)^2-4)}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{-99-25(x-2)^2+100}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{1-25(x-2)^2}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{1-(5x-10)^2}}\mathrm{d}x$

$=\frac{1}{5}\arcsin(5x-10)+c$

47) a)

$-27 + 24x - 4x^2$

$\equiv -27 - 4(x^2 - 6x)$

$\equiv -27 - 4((x-3)^2 - 9)$

$\equiv -27 - 4(x-3)^2 + 36$

$\equiv 9 - 4(x-3)^2$

47) b)

Let $\frac{2}{3}x - 2 = \sin u \Rightarrow x = \frac{3}{2}\sin u + 3 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\cos u \Rightarrow \mathrm{d}x = \frac{3}{2}\cos u \ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{-27+24x-4x^2}}\mathrm{d}x = \int \frac{1}{\sqrt{9-4(x-3)^2}}\mathrm{d}x = \int \frac{1}{\sqrt{9\left(1-\frac{4}{9}(x-3)^2\right)}}\mathrm{d}x = \frac{1}{3}\int \frac{1}{\sqrt{1-\left(\frac{2}{3}x-2\right)^2}}\mathrm{d}x$

$= \frac{1}{3}\int \frac{1}{\sqrt{1-\sin^2 u}} \times \frac{3}{2}\cos u \, \mathrm{d}u$

$= \frac{1}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\mathrm{d}x$

$= \frac{1}{2}\int \frac{\cos u}{\cos u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\arcsin\left(\frac{2}{3}x - 2\right) + c$

48) a)

$\int \frac{1}{\sqrt{-11-36x-9x^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{-11-9(x^2+4x)}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{-11-9((x+2)^2-4)}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{-11-9(x+2)^2+36}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{25-9(x+2)^2}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{25\left(1-\frac{9}{25}(x+2)^2\right)}}\mathrm{d}x$

$= \frac{1}{5}\int \frac{1}{\sqrt{1-\left(\frac{3}{5}x+\frac{6}{5}\right)^2}}\mathrm{d}x$

$= \frac{1}{5} \times \frac{1}{\frac{3}{5}} \times \arcsin\left(\frac{3}{5}x + \frac{6}{5}\right) + c$

$= \frac{1}{3}\arcsin\left(\frac{3}{5}x + \frac{6}{5}\right) + c$

48) b)

$$\int \frac{1}{\sqrt{3+10x-25x^2}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{3-25\left(x^2-\frac{2}{5}x\right)}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{3-25\left(\left(x-\frac{1}{5}\right)^2-\frac{1}{25}\right)}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{3-25\left(x-\frac{1}{5}\right)^2+1}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{4-25\left(x-\frac{1}{5}\right)^2}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{4\left(1-\frac{25}{4}\left(x-\frac{1}{5}\right)^2\right)}}\,\mathrm{d}x$$

$$= \frac{1}{2}\int \frac{1}{\sqrt{1-\left(\frac{5}{2}x-\frac{1}{2}\right)^2}}\,\mathrm{d}x$$

$$= \frac{1}{2} \times \frac{1}{\frac{5}{2}} \times \arcsin\left(\frac{5}{2}x-\frac{1}{2}\right)+c$$

$$= \frac{1}{5}\arcsin\left(\frac{5}{2}x-\frac{1}{2}\right)+c$$

49)

$$\int_2^{\frac{9}{2}} \frac{1}{\sqrt{21+4x-x^2}}\,\mathrm{d}x$$

$$= \int_2^{\frac{9}{2}} \frac{1}{\sqrt{21-(x^2-4x)}}\,\mathrm{d}x$$

$$= \int_2^{\frac{9}{2}} \frac{1}{\sqrt{21-((x-2)^2-4)}}\,\mathrm{d}x$$

$$= \int_2^{\frac{9}{2}} \frac{1}{\sqrt{21-(x-2)^2+4}}\,\mathrm{d}x$$

$$= \int_2^{\frac{9}{2}} \frac{1}{\sqrt{25-(x-2)^2}}\,\mathrm{d}x$$

$$= \int_2^{\frac{9}{2}} \frac{1}{\sqrt{25\left(1-\frac{1}{25}(x-2)^2\right)}}\,\mathrm{d}x$$

$$= \frac{1}{5}\int_2^{\frac{9}{2}} \frac{1}{\sqrt{1-\left(\frac{x-2}{5}\right)^2}}\,\mathrm{d}x$$

$$= \left[\frac{1}{5} \times \frac{1}{\frac{1}{5}} \arcsin\left(\frac{x-2}{5}\right)\right]_2^{\frac{9}{2}}$$

$$= \left[\arcsin\left(\frac{x-2}{5}\right)\right]_2^{\frac{9}{2}}$$

$$= \left(\arcsin\frac{1}{2}\right) - (\arcsin 0)$$

$$= \frac{\pi}{6} - 0$$

$$= \frac{\pi}{6}$$

So $k = \frac{1}{6}$

50)

Volume

$= \pi \int_0^{20} \left(\frac{1}{\sqrt{400+x^2}}\right)^2 \mathrm{d}x$

$= \pi \int_0^{20} \frac{1}{400+x^2} \mathrm{d}x$

$= \pi \int_0^{20} \frac{1}{400\left(1+\frac{1}{400}x^2\right)} \mathrm{d}x$

$= \frac{\pi}{400} \int_0^{20} \frac{1}{1+\left(\frac{x}{20}\right)^2} \mathrm{d}x$

$= \frac{\pi}{400} \left[20 \arctan\left(\frac{x}{20}\right)\right]_0^{20}$

$= \frac{\pi}{400} (20 \arctan 1 - 20 \arctan 0)$

$= \frac{\pi}{400} (5\pi - 0)$

$= \frac{\pi^2}{80}$ units^3

51)

$\int_{\frac{2}{5}}^{\infty} \frac{1}{4+25x^2} \mathrm{d}x = \lim_{a\to\infty} \int_{\frac{2}{5}}^{a} \frac{1}{4+25x^2} \mathrm{d}x$

$= \lim_{a\to\infty} \int_{\frac{2}{5}}^{a} \frac{1}{4\left(1+\frac{25}{4}x^2\right)} \mathrm{d}x$

$= \lim_{a\to\infty} \int_{\frac{2}{5}}^{a} \frac{1}{4\left(1+\left(\frac{5}{2}x\right)^2\right)} \mathrm{d}x$

$= \lim_{a\to\infty} \left[\frac{1}{4} \times \frac{1}{\frac{5}{2}} \times \arctan\left(\frac{5}{2}x\right)\right]_{\frac{2}{5}}^{a}$

$= \lim_{a\to\infty} \left[\frac{1}{10} \arctan\left(\frac{5}{2}x\right)\right]_{\frac{2}{5}}^{a}$

$= \lim_{a\to\infty} \left(\frac{1}{10} \arctan\left(\frac{5}{2}a\right) - \frac{1}{10} \arctan 1\right)$

As $a \to \infty$, $\arctan\left(\frac{5}{2}a\right) \to \frac{\pi}{2}$

$\therefore \lim_{a\to\infty} \left(\frac{1}{10} \arctan\left(\frac{5}{2}a\right) - \frac{1}{10} \arctan 1\right) = \frac{1}{10} \times \frac{\pi}{2} - \frac{1}{10} \times \frac{\pi}{4}$

$= \frac{\pi}{20} - \frac{\pi}{40}$

$= \frac{\pi}{40}$

So $k = \frac{1}{40}$